order to be successful. Panchayats therefore were looked upon as instruments of decentralisation and participatory democracy. Our national movement was concerned about the enormous concentration of powers in the hands of the Governor General sitting at Delhi. Therefore, for our leaders, independence meant an assurance that there will be decentralisation of decision making, executive and administrative powers.

> The independence of India should mean the independence of the whole of India...Independence must begin at the bottom. Thus every village will be a republic... It follows therefore that every village has to be self-sustained and capable of managing its affairs. In this structure composed of innumerable villages, there will be ever-widening, ever-ascending circles. Life will be a pyramid with the apex sustained by the bottom - Mahatma Gandhi

When the Constitution was prepared, the subject of local government was assigned to the States. It was also mentioned in the Directive Principles as one of the policy directives to all governments in the country. As you have read in Chapter 2, being a part of the Directive Principles of State Policy, this provision of the Constitution was non-justiciable and primarily advisory in its nature.

It is felt that the subject of local government including panchayats did not receive adequate importance in the Constitution. Do you know why this happened? A few reasons can be advanced here. Firstly, the turmoil due to the Partition resulted in a strong unitary inclination in the Constitution. Nehru himself looked upon extreme localism as a threat to unity and integration of the nation. Secondly, there was a powerful voice in the Constituent Assembly led by Dr. B.R. Ambedkar which felt that the faction and caste-ridden nature of rural society would defeat the noble purpose of local government at the rural level.

However, nobody denied the importance of people's participation in development planning. Many members of the Constituent Assembly wanted Village Panchayats to be the basis of democracy

Reprint 2026-27

in India but they were concerned about factionalism and many other ills present in the villages.

"... in the interests of democracy, the villages maybe trained in the art of self-government, even autonomy... We must be able to reform the villages and introduce democratic principles of government there..."



Ananthasayanam Ayyangar, CAD, Vol. VII, p. 428, 17 November 1948

Local Governments in Independent India

Local governments got a fillip after the 73rd and 74th Constitution Amendment Acts. But even before that, some efforts in the direction of developing local government bodies had already taken place. First in the line was the Community Development Programme in 1952, which sought to promote people's participation in local development in a range of activities. In this background, a three-tier Panchayati Raj system of local government was recommended for the rural areas. Some States (like Gujarat, Maharashtra) adopted the system of elected local bodies around 1960. But in many States those local bodies did not have enough powers and functions to look after the local development. They were very much dependent on the State and central governments for financial assistance. Many States did not think it necessary to establish elected local bodies. In many instances, local bodies were dissolved and the local government was handed over to government officers. Many States had indirect elections to most local bodies. In many States, elections to the local bodies were postponed from time to time.



Why are people afraid of factionalism at the village level when all the political parties and organisations or even my class has factions? Are groups and factions always so bad?

After 1987, a thorough review of the functioning of local government institutions was initiated. In 1989 the P.K.Thungon Committee recommended constitutional recognition for the local government bodies. A constitutional amendment to provide for periodic elections to local government institutions, and enlistment of appropriate functions to them, along with funds, was recommended.

Check your progress

- Both Nehru and Dr. Ambedkar were not very enthusiastic about local government bodies. Did they have similar objections to local governments?
- What was the constitutional provision about local governments before 1992?
- Which were the States that had established local government during the 1960s and 1970s ?

73RD AND 74TH AMENDMENTS

In 1989, the central government introduced two constitutional amendments. These amendments aimed at strengthening local governments and ensuring an element of uniformity in their structure and functioning across the country.

The Constitution of Brazil has created States, Federal Districts and Municipal Councils. Each of these is assigned independent powers and jurisdiction. Just as the Republic cannot interfere in the affairs of the States (except on grounds provided by the constitution), states are prohibited from interfering in the affairs of the municipal councils. This provision protects the powers of the local government.

Later in 1992, the 73rd and 74th constitutional amendments were passed by the Parliament. The 73rd Amendment is about rural local governments (which are also known as Panchayati Raj Institutions

Reprint 2026-27

or PRIs) and the 74th amendment made the provisions relating to urban local government (Nagarpalikas). The 73rd and 74th Amendments came into force in 1993.

We have noticed earlier that local government is a 'State subject'. States are free to make their own laws on this subject. But once the Constitution was amended, the States had to change their laws about local bodies in order to bring these in conformity with the amended Constitution. They were given one year's time for making necessary changes in their respective State laws in the light of these amendments.



If I understand this correctly, the centre forced local government reforms on the States. This is funny: you adopt decentralisation through a centralised process!

73rd Amendment

Let us now examine the changes brought about by the 73rd amendment in Panchayati Raj institutions.

Three Tier Structure

All States now have a uniform three tier Panchayati Raj structure. At the base is the 'Gram Panchayat'. A Gram Panchayat covers a village or group of villages. The intermediary level is the Mandal (also referred to as Block or Taluka). These bodies are called Mandal or Taluka Panchayats. The intermediary level body need not be constituted in smaller States. At the apex is the Zilla Panchayat covering the entire rural area of the District.

The amendment also made a provision for the mandatory creation of the Gram Sabha. The Gram Sabha would comprise all the adult members registered as voters in the Panchayat area. Its role and functions are decided by State legislation.



Does a Gram sabha mean the democratic forum of the entire village? Do Gram sabhas actually meet regularly?

Elections

All the three levels of Panchayati Raj institutions are elected directly by the people. The term of each Panchayat body is five years. If the State government dissolves the Panchayat before the end of its five year term,

fresh elections must be held within six months of such dissolution. This is an important provision that ensures the existence of elected local bodies. Before the 73rd amendment, in many States, there used to be indirect elections to the district bodies and there was no provision for immediate elections after dissolution.

Reservations

One third of the positions in all panchayat institutions are reserved for women. Reservations for Scheduled Castes and Scheduled Tribes are also provided for at all the three levels, in proportion to their population. If the States find it necessary, they can also provide for reservations for the other backward classes (OBCs).



We read in the chapter on Elections that the bill for reservations for women in the Assemblies and in the Parliament could not be passed. How come women's reservations in local bodies were accepted so easily?

It is important to note that these reservations apply not merely to ordinary members in Panchayats but also to the positions of Chairpersons or '*Adhyakshas*' at all the three levels. Further, reservation of one-third of the seats for women is not merely in the general category of seats but also within the seats reserved for Scheduled Castes, Scheduled Tribes and backward castes. This means that a seat may be reserved simultaneously for a woman candidate and one belonging to the Scheduled Castes or Scheduled Tribes. Thus, a Sarpanch would have to be a Dalit woman or an Adivasi woman.

Transfer of Subjects

Twenty-nine subjects, which were earlier in the State list of subjects, are identified and listed in the Eleventh Schedule of the Constitution. These subjects are to be transferred to the Panchayati Raj institutions. These subjects were mostly linked to development and welfare functions at the local level. The actual transfer of these functions depends upon the State legislation. Each State decides how many of these twenty-nine subjects would be transferred to the local bodies.

© NCERT not to be republished

Article 243G. Powers, authority and responsibilities of Panchayats.—....., the Legislature of a State may, by law, endow the Panchayats with such powers and authority..... ...with respect to—.....the matters listed in the Eleventh Schedule.

Some subjects listed in the eleventh schedule

1. Agriculture, ...
3. Minor irrigation, water management and watershed development.

....

8. Small scale industries, including food processing industries.

.....

10. Rural housing.
11. Drinking water.

.....

13. Roads, culverts,.....
14. Rural electrification,.....

.....

16. Poverty alleviation programme.
17. Education, including primary and secondary schools.
18. Technical training and vocational education.
19. Adult and non-formal education.
20. Libraries.
21. Cultural activities.
22. Markets and fairs.
23. Health and sanitation, including hospitals, primary health centres and dispensaries.
24. Family welfare.
25. Women and child development.
26. Social welfare, ...
27. Welfare of the weaker sections, and in particular, of the Scheduled Castes and the Scheduled Tribes.
28. Public distribution system.

.....



Why are subjects only from State list to be transferred? Why can't we transfer some subjects from the Union List also?

© NCERT not to be republished

The provisions of the 73rd amendment were not made applicable to the areas inhabited by the Adivasi populations in many States of India. In 1996, a separate act was passed extending the provisions of the Panchayat system to these areas. Many Adivasi communities have their traditional customs of managing common resources such as forests and small water reservoirs, etc. Therefore, the new act protects the rights of these communities to manage their resources in ways acceptable to them. For this purpose, more powers are given to the Gram Sabhas of these areas and elected village panchayats have to get the consent of the Gram Sabha in many respects. The idea behind this act is that local traditions of self government should be protected while introducing modern elected bodies. This is only consistent with the spirit of diversity and decentralisation.

State Election Commissioners

The State government is required to appoint a State Election Commissioner who would be responsible for conducting elections to the Panchayati Raj institutions. Earlier, this task was performed by the State administration which was under the control of the State government. Now, the office of the State Election Commissioner is autonomous like the Election Commissioner of India. However, the State Election Commissioner is an independent officer and is not linked to nor is this officer under the control of the Election Commission of India.



State governments themselves are poor. In the last chapter we read that they ask for money from the Central government. How can they give money to the local government?

State Finance Commission

The State government is also required to appoint a State Finance Commission once in five years. This Commission would examine the financial position of the local governments in the State. It would also review the distribution of revenues between the State and local governments on the one hand and between rural and urban local governments on the other. This innovation ensures that allocation of funds to the rural local governments will not be a political matter.

© NCERT not to be republished

Activity

- Identify some of the powers that your State government has delegated to panchayats.

74th Amendment

As we mentioned earlier, the 74th amendment dealt with urban local bodies or Nagarpalikas.

What is an urban area? It is very easy to identify a big city like Mumbai or Kolkata, but it is not so easy to say this about some very small urban areas that are somewhere between a village and a town. The Census of India defines an urban area as having: (i) a minimum population of 5,000; (ii) at least 75 per cent of male working population engaged in non-agricultural occupations and (iii) a density of population of at least 400 persons per sq. km. As per the 2011 Census, about 31% of India's population lives in urban areas.

In many ways the 74th amendment is a repetition of the 73rd amendment, except that it applies to urban areas. All the provisions of the 73rd amendment relating to direct elections, reservations, transfer of subjects, State Election Commission and State Finance Commission are incorporated in the 74th amendment also and thus apply to Nagarpalikas. The Constitution also mandated the transfer of a list of functions from the State government to the urban local bodies. These functions have been listed in the Twelfth Schedule of the Constitution.



Can I hope that these urban local bodies will do something for better housing for the slum dwellers? Or at least provide them toilets?

Implementation of 73rd and 74th Amendments

All States have now passed a legislation to implement the provisions of the 73rd and 74th amendments. During the ten years since these amendments came into force (1994-2004) most States have had at least two rounds of elections to the local bodies. States like Madhya Pradesh, Rajasthan and a few others have in fact held three elections so far.

© NCERT not to be republished

READ AN IMAGE



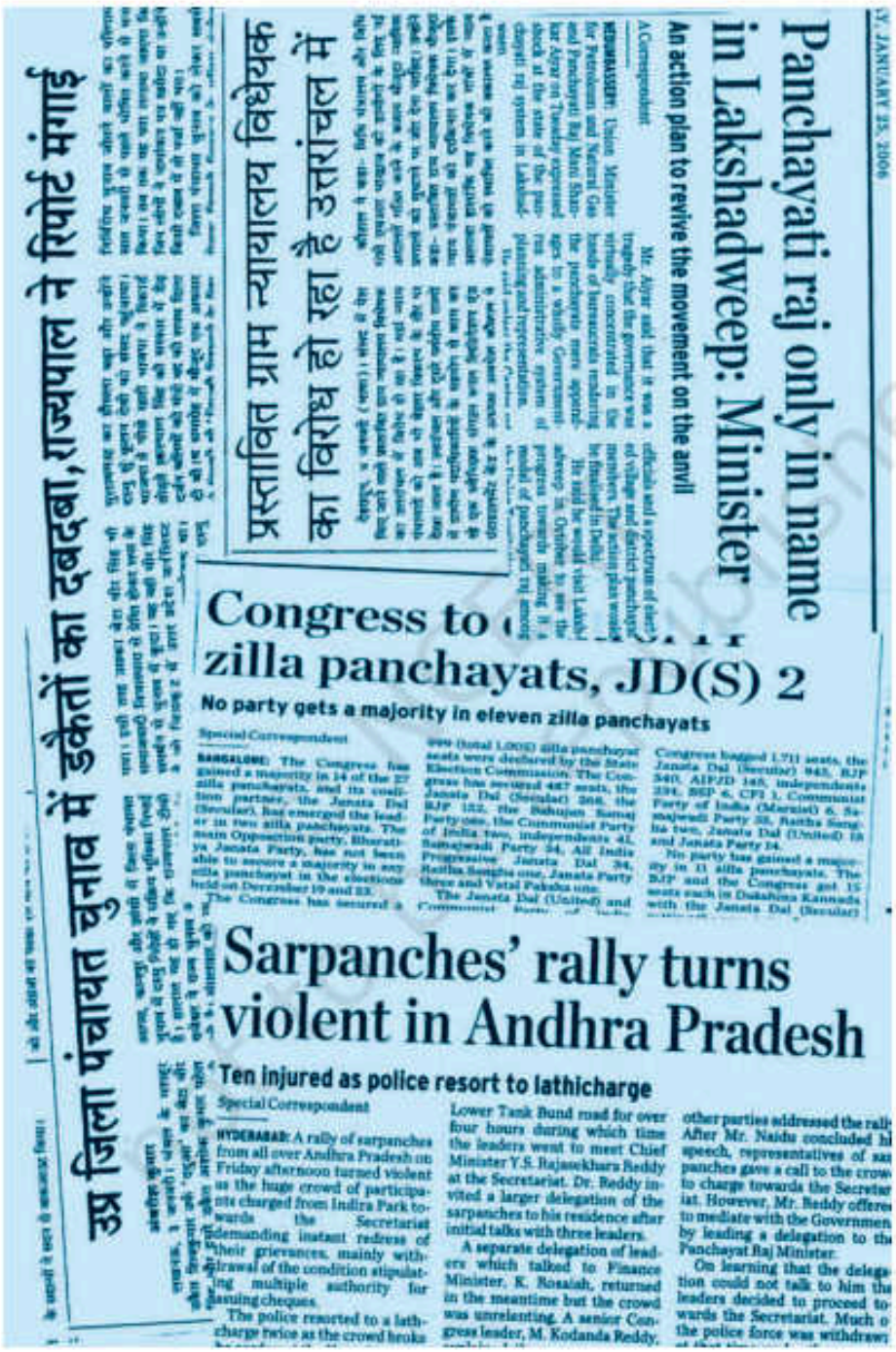
This flag is a symbol of the expectations of the people about local governments. People don't want only formal laws. They want genuine implementation of those laws. Write briefly what you think about this slogan — We are the government here in the village!

Today there are more than 600 Zilla Panchayats, about 6,000 block or intermediary Panchayats, and 2,40,000 Gram Panchayats in rural India and over 100 city Corporations, 1400 town Municipalities and over 2000 Nagar Panchayats in urban India. More than 32 lakh members are elected to these bodies every five years. Of these, at least 13 lakhs are women. In the State Assemblies and Parliament put together we have less than 5000 elected representatives. With local bodies, the number of elected representatives has increased significantly.

The 73rd and 74th amendments have created uniformity in the structures of Panchayati Raj and Nagarpalika institutions across the country. The presence of these local institutions is by itself a significant achievement and would create an atmosphere and platform for people's participation in government.

The provision for reservation for women at the Panchayats and Nagarpalikas has ensured the presence of a significant number of women in local bodies. As this reservation is also applicable for the positions of Sarpanch and Adhyaksha, a large number of women elected representatives have come to occupy these positions. There are at least 200 women Adhyakshas in Zilla Panchayats, another 2000 women who are Presidents of the block or taluka panchayats and more than 80,000 women Sarpanchas in Gram Panchayats.

Reprint 2026-27



उप जिला पंचायत चुनाव में डकैतों का दबदबा, राज्यपाल ने रिपोर्ट मंगाई

प्रस्तावित ग्राम न्यायालय विधेयक का विरोध हो रहा है उत्तरांचल में

Panchayati raj only in name in Lakshadweep: Minister

An action plan to revive the movement on the anvil

Congress to ... zilla panchayats, JD(S) 2

No party gets a majority in eleven zilla panchayats

Sarpanches' rally turns violent in Andhra Pradesh

Ten injured as police resort to lathicharge

© NCERT not to be republished

We also have more than 30 women Mayors in Corporations, over 500 women Adhyakshas of Town Municipalities and nearly 650 Nagar Panchayats headed by women. Women have gained more power and confidence by asserting control over resources. Their presence in these institutions has given many women a greater understanding of the working of politics. In many cases, they have brought a new perspective and a greater sensitivity to discussions at local bodies. In many cases, women were unable to assert their presence or were mere proxies for the male members of their family who sponsored their election. Such instances, however are becoming fewer.

READ AN IMAGE



Look at this photograph. The local Sarkar is sitting out in the sun. Is there any other feature that strikes you?

While reservations for Scheduled Castes and Tribes are mandated by the constitutional amendment, most States have also made a provision to reserve seats for Backward Castes. As the Indian population has 16.2 per cent Scheduled Castes and 8.2 per cent Scheduled Tribes, about 6.6 lakh elected members in the urban and local bodies hail from these two communities. This has

Reprint 2026-27

significantly altered the social profile of local bodies. These bodies have thus become more representative of the social reality they operate within. Sometimes this leads to tensions. The dominant social groups which controlled the village earlier do not wish to give up their power. This leads to intensification of struggle for power. But tension and struggle is not always bad. Whenever there is an attempt to make democracy more meaningful and give power to those who did not enjoy it earlier, there is bound to be some conflict and tension in society.

The Constitutional amendments assigned as many as 29 subjects to the local governments. All these subjects are related to functions linked to local welfare and development needs. The experience with the functioning of local government in the past decade has shown that local governments in India enjoy limited autonomy to perform the functions assigned to them. Many States have not transferred most of the subjects to the local bodies. This means that the local bodies cannot really function in an effective manner. Therefore, the entire exercise of electing so many representatives becomes somewhat symbolic. Some people criticise the formation of the local bodies because this has not changed the way in which decisions are taken at the central and the State level. People at the local level do not enjoy much powers of choosing welfare programmes or allocation of resources.



So, the law is good but it is mostly on paper. Is this what they call the gap between theory and practice?

Bolivia is frequently cited as one of the most successful cases of democratic decentralisation in Latin America. In 1994, the Popular Participation Law decentralised power to the local level, allowing for the popular election of mayors, dividing the country into municipalities, and crafting a system of automatic fiscal transfers to the new municipalities. Bolivia is divided into 314 municipal governments. These governments in Bolivia are headed by popularly-elected mayors (*presidente municipal*) and a municipal council (*cabildo*). Local elections occur nationwide every five years.

Bolivian local governments have been entrusted with building local health and education facilities, as well as maintenance of this infrastructure. In Bolivia, 20% of nationwide tax collections are distributed among municipalities on a per capita basis. While these municipalities may levy taxes on motor vehicles, urban property, and large agricultural properties, fiscal transfers provide the bulk of the operating budget for these units.

Local bodies have very little funds of their own. The dependence of local bodies on the State and central governments for financial support has greatly eroded their capacity to operate effectively. While rural local bodies raise 0.24% of the total revenues collected, they account for 4% of the total expenditure made by the government. So they earn much less than they spend. That makes them dependent on those who give them grants.

Conclusion

This experience suggests that local governments continue to be agencies implementing the welfare and development schemes of the central and State government. Giving more power to local government means that we should be prepared for real decentralisation of power. Ultimately, democracy means that power should be shared by the people; people in the villages and urban localities must have the power to decide what policies and programmes they want to adopt. As you have studied earlier, democracy means decentralisation of power and giving more and more power to the people. The laws about local governments are an important step in the direction of democratisation. But the true test of democracy is not merely in the legal provisions but in the practice of those provisions.

Reprint 2026-27

There are many articles in the Constitution, which mention that these articles can be amended by a simple law of the Parliament. No special procedure for amendment is required in such cases and there is no difference at all between an amendment and an ordinary law. These parts of the Constitution are very flexible. Read carefully the following text of some articles of the Constitution. In both these articles, the wording '*by law*' indicates that these articles can be modified by the Parliament without recourse to the procedure laid down in Article 368. Many other articles of the Constitution can be modified by the Parliament in this simple manner.

Article 2: Parliament may by law admit into the union .....new states....

Article 3: Parliament may by law... b) increase the area of any state....

For amending the remaining parts of the Constitution, provision has been made in Article 368 of the Constitution. In this article, there are two methods of amending the Constitution and they apply to two different sets of articles of the Constitution. One method is that amendment can be made by special majority of the two houses of the Parliament. The other method is more difficult: it requires special majority of the Parliament and consent of half of the State legislatures. Note that all amendments to the Constitution are initiated only in the Parliament. Besides the special majority in the Parliament no outside agency——like a constitution commission or a separate body—is required for amending the Constitution.

Similarly, after the passage in the Parliament and in some cases, in State legislatures, no referendum is required for ratification of the amendment. An amendment



What happens if some States want an amendment to the Constitution? Can't they propose an amendment? I think this is another example of favouring the centre against the States!

© NCERT not to be republished

bill, like all other bills, goes to the President for his assent, but in this case, the President has no powers to send it back for reconsideration. These details show how rigid and complicated the amending process could have been. Our Constitution avoids these complications. This makes the amendment procedure relatively simple. But more importantly, this process underlines an important principle: only elected representatives of the people are empowered to consider and take final decisions on the question of amendments. Thus, sovereignty of elected representatives (parliamentary sovereignty) is the basis of the amendment procedure.

Special Majority

In the chapters on Election, Executive and Judiciary, we have come across provisions that require 'special majority'. Let us repeat again what special majority means. Ordinarily, all business of the legislature requires that a motion or resolution or bill should get the support of a simple majority of the members voting at that time. Suppose that at the time of voting on a bill, 247 members were present in the house and all of them participated in the voting on the bill. Then, the bill would be passed if at least 124 members voted in favour of the bill. Not so in the case of an amendment bill. Amendment to the Constitution requires two different kinds of special majorities: in the first place, those voting in favour of the amendment bill should constitute at least half of the total strength of that House. Secondly, the supporters of the amendment bill must also constitute two-thirds of those who actually take part in voting. Both Houses of the Parliament must pass the amendment bill separately in this same manner (there is no provision for a joint session). For every amendment bill, this special majority is required.

Can you see the significance of this requirement? In the Lok Sabha there are 545 members. Therefore, any amendment must be supported by a minimum of 273 members. Even if only 300 members are present at the time of voting, the amendment bill must get the support of 273 out of them. But imagine that 400 members of Lok Sabha have voted on an amendment bill. How many members should support the bill to get the bill passed?

In addition to this, both the Houses must pass the amendment bill (with special majorities) separately. This means that unless there

Two principles dominate the various procedures of amending the constitutions in most modern constitutions.

- One is the principle of special majority. For instance, the constitutions of U.S., South Africa, Russia, etc. have employed this principle: In the case of constitution of US, it is two-thirds majority, while in South Africa and Russia, for some amendments, three-fourths majority is required.
- The other principle that is popular among many modern constitutions is that of people's participation in the process of amending the constitution. In Switzerland, people can even initiate an amendment. Other examples of countries where people initiate or approve amendment to the constitution are Russia and Italy, among others.



I am fed up with this business of special majority. It forces you to make difficult calculations all the time. Is it politics or maths?

is sufficient consensus over the proposed amendment, it cannot be passed. If the party in power enjoys very thin majority, it can pass legislation of its choice and can get budget approved even if the opposition does not agree. But it would need to take at least some opposition parties into confidence, if it wanted to amend the Constitution. So, the basic principle behind the amending procedure is

"If those who are dissatisfied with the Constitution have only to obtain a 2/3 majority and if they cannot obtain even (that)..., their dissatisfaction with the Constitution cannot be deemed to be shared by the general public."

Note that Dr. Ambedkar is talking here not only of parliamentary majority. He refers to 'sharing (of the views) by the general public'. This indicates that behind the majority there is the principle of public opinion that governs decision-making.

Dr. Ambedkar, CAD, Vol. XI, p. 976, 25 November 1949



© NCERT not to be republished

that it should be based on broad support among the political parties and parliamentarians.

Ratification by States

For some articles of the Constitution, special majority is not sufficient. When an amendment aims to modify an article related to distribution of powers between the States and the central government, or articles related to representation, it is necessary that the States must be consulted and that they give their consent. We have studied the federal nature of the Constitution. Federalism means that powers of the States must not be at the mercy of the central government. The Constitution has ensured this by providing that legislatures of half the States have to pass the amendment bill before the amendment comes into effect. We can say that for some parts of the Constitution, greater or wider consensus in the polity is expected. This provision also respects the States and gives them participation in the process of amendment. At the same time, care is taken to keep this procedure somewhat flexible even in its more rigid format: consent of only half the States is required and simple majority of the State legislature is sufficient. Thus, the amendment process is not impracticable even after taking into consideration this more stringent condition.

We may summarise that the Constitution of India can be amended through large-scale consensus and limited participation of the States. The founding fathers took care that Constitution would not be open to easy tampering. And yet, future generations were given the right to amend and modify according to the needs and requirements of the time.

Check your progress

For making the following amendments to the Constitution of India, what conditions need to be fulfilled? Place a tick mark in the chart wherever applicable.

Reprint 2026-27

Subject of amendment	Special majority	Ratification by States
Citizenship clause		
Right to freedom of religion		
Changes in the Union List		
Changes in State boundaries		
Provision regarding Election Commission		

Why Have There Been so Many Amendments?

On 26 January 2024, the Constitution of India completed 74 years of its existence. In these years, it has been amended 106 times. Given the relatively difficult method of amending the Constitution, the number of amendments appears quite high. Let us try to find out how it is that so many amendments took place and what it means.

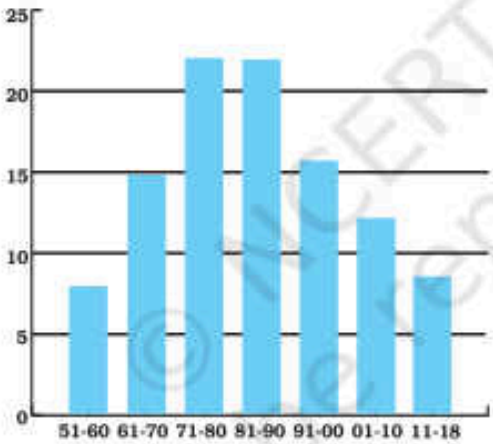
Let us first look at the brief history of the amendments: look carefully at the graphs below. The same information is presented in two different ways. The first graph depicts the number of constitution amendments made every ten years; the bar indicates the number of amendments in that period. The second graph depicts the time taken for every ten amendments; the bar depicts the years taken for ten amendments. You will notice that the two decades from 1970 to 1990 saw a large number of amendments. On the other hand, the second graph tells one more story: ten amendments took place between a short span of three



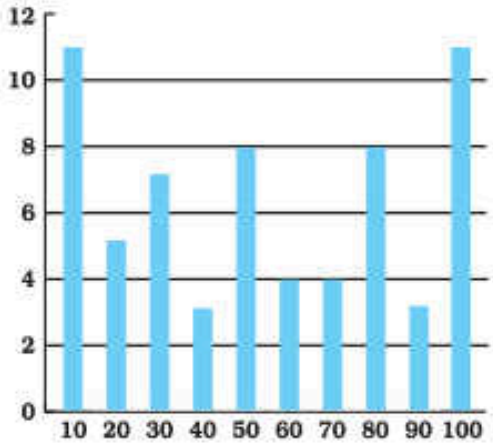
Why was our Constitution amended so many times? Is there something wrong with our society or with the Constitution?

© NCERT not to be republished

years between 1974 and 1976. And again, in just three years, from 2001 to 2003, ten amendments took place. In the political history of our country, these two periods are remarkably different. The first was a period of Congress domination. Congress party had a vast majority in Parliament (it had 352 seats in the Lok Sabha and a majority in most State Assemblies). On the other hand, the period between 2001 and 2003 was a period marked by coalition politics. It was also a period when different parties were in power in different States. The bitter rivalry between the BJP and its opponents is another feature of this period. And yet, this period saw as many as ten amendments in just three years. So, the incidence of amendments is not dependent merely on the nature of majority of the ruling party alone.



Graph 1
Amendments per decade



Graph 2
Years taken for every ten amendments

© NCERT not to be republished

as attempts by the ruling party to subvert the Constitution. In particular, the 38th, 39th and 42nd amendments have been the most controversial amendments so far. These three amendments were made in the background of internal emergency declared in the country from June 1975. They sought to make basic changes in many crucial parts of the Constitution.

The 42nd amendment was particularly seen as a wide-ranging amendment affecting large parts of the Constitution. It was also an attempt to override the ruling of the Supreme Court given in the Kesavananda case. Even the duration of the Lok Sabha was extended from five to six years. In the chapter on Rights, you have read about Fundamental Duties. They were included in the Constitution by this amendment act. The 42nd amendment also put restrictions on the review powers of the Judiciary. It was said at that time that this amendment was practically a rewriting of many parts of the original Constitution. Do you know that this amendment made changes to the Preamble, to the seventh schedule of the Constitution and to 53 articles of the Constitution? Many MPs belonging to the opposition parties were in jail when this amendment was passed in Parliament. In this backdrop, elections were held in 1977 and the ruling party (Congress) was defeated. The new government thought it necessary to reconsider these controversial amendments and through the 43rd and 44th amendments, cancelled most of the changes that were effected by the 38th, 39th and the 42nd amendments. The constitutional balance was restored by these amendments.



So, it is all about politics! Didn't I say that this entire thing about constitutions and amendments is linked to politics rather than law?



Activity

Find out the amendments about the right to education (RTE) and the Goods and Services Tax (GST). What do you think is the importance of these amendments?

Reprint 2026-27

Basic Structure and Evolution of the Constitution

One thing that has had a long lasting effect on the evolution of the Indian Constitution is the theory of the basic structure of the Constitution. You know already that the Judiciary advanced this theory in the famous case of Kesavananda Bharati. This ruling has contributed to the evolution of the Constitution in the following ways:

- It has set specific limits to Parliament's power to amend the Constitution. It says that no amendment can violate the basic structure of the Constitution;
- It allows Parliament to amend any and all parts of the Constitution (within this limitation); and
- It places the Judiciary as the final authority in deciding if an amendment violates basic structure and what constitutes the basic structure.

The Supreme Court gave the Kesavananda ruling in 1973. In the past four decades, this decision has governed all interpretations of the Constitution and all institutions in the country have accepted the theory of basic structure. In fact, the theory of basic structure is itself an example of a living constitution. There is no mention of this theory in the Constitution. It has emerged from judicial interpretation. Thus, the Judiciary and its interpretation have practically amended the Constitution without a formal amendment.

All living documents evolve in this manner through debates, arguments, competition and practical politics. Since 1973, the Court has, in many cases, elaborated upon this theory of basic structure and given instances of what constitutes the basic structure of the Constitution of India. In a sense, the basic structure doctrine has further consolidated the balance between rigidity and flexibility: by saying that certain parts cannot be amended, it has underlined the rigid nature while by allowing amendments to all others it has underlined the flexible nature of the amending process.



Ah! So it is the judiciary that has the final word! Is this also judicial activism?

© NCERT not to be republished

Review of the Constitution

In the late nineties, efforts were made to review the entire Constitution. In the year 2000 a commission to review the working of the Constitution was appointed by the Government of India under the chairmanship of a retired Chief Justice of the Supreme Court, Justice Venkatachaliah. Opposition parties and many other organisations boycotted the commission. While a lot of political controversy surrounded this commission, the commission stuck to the theory of basic structure and did not suggest any measures that would endanger the basic structure of the Constitution. This shows the significance of the basic structure doctrine in our constitutional practice.

There are many other examples of how judicial interpretation changed our understanding of the Constitution. In many decisions the Supreme Court had held that reservations in jobs and educational institutions cannot exceed fifty per cent of the total seats. This has now become an accepted principle. Similarly, in the case involving reservations for other backward classes, the Supreme Court introduced the idea of creamy layer and ruled that persons belonging to this category were not entitled to benefits under reservations. In the same manner, the Judiciary has contributed to an informal amendment by interpreting various provisions concerning right to education, right to life and liberty and the right to form and manage minority educational institutions. These are instances of how rulings by the Court contribute to the evolution of the Constitution.



It's all wrong. First they say that an amendment requires consensus and now we see that Judges change the whole meaning of the Constitution.

© NCERT not to be republished

Check your progress

State whether the following statements are correct or not:

- After the Basic Structure ruling, Parliament does not have power to amend the Constitution.
- The Supreme Court has given a clear list of the basic features of our Constitution, which cannot be amended.
- Judiciary has the power to decide whether an amendment violates basic structure or not.
- The Kesavananda Bharati ruling has set clear limits on Parliament's power to amend the Constitution.

Constitution as a Living Document

We have described our Constitution as a living document. What does that mean?

Almost like a living being, this document keeps responding to the situations and circumstances arising from time to time. Like a living being, the Constitution responds to experience. In fact that is the answer to the riddle we mentioned at the beginning about the durability of the Constitution. Even after so many changes in the society, the Constitution continues to work effectively because of this ability to be dynamic, to be open to interpretations and the ability to respond to the changing situation. This is a hallmark of a democratic constitution. In a democracy, practices and ideas keep evolving over time and the society engages in experiments according to these. A constitution, which protects democracy and yet allows for evolution of new practices becomes not only durable but also the object of respect from the citizens. The important point is: has the Constitution been able to protect itself and protect democracy?

In the past six decades, some very critical situations arose in the politics and constitutional development of the country. We have made a brief reference to some of these in this chapter already. In terms of constitutional-legal issues, the most serious question that came up

© NCERT not to be republished

again and again from 1950 was about the supremacy of the Parliament. In a parliamentary democracy, the Parliament represents the people and therefore, it is expected to have an upper hand over both Executive and Judiciary. At the same time, there is the text of the Constitution and it has given powers to other organs of the government. Therefore, the supremacy of the Parliament has to operate within this framework. Democracy is not only about votes and people's representation. It is also about the principle of rule of law. Democracy is also about developing institutions and working through these institutions. All the political institutions must be responsible to the people and maintain a balance with each other.



I get it! It's like a see-saw. Or is it a game of tug of war?

Contribution of the Judiciary

During the controversy between the Judiciary and the Parliament, the Parliament thought that it had the power and responsibility to make laws (and amendments) for furthering the interests of the poor, backward and the needy. The Judiciary insisted that all this has to take place within the framework provided by the Constitution and pro-people measures should not bypass legal procedures, because, once you bypass laws even with good intentions, that can give an excuse to the power holders to use their power arbitrarily. And democracy is as much about checks on arbitrary use of power as it is about the well-being of the people.

The success of the working of the Indian Constitution lies in resolving these tensions. The Judiciary, in its famous Kesavananda ruling found a way out of the existing complications by turning to the spirit of the Constitution rather than its letter. If you read the Constitution, you will not find any mention of the 'basic structure' of the Constitution. Nowhere does the Constitution say that such and such are part of the basic structure. In this sense, the 'basic structure' theory is the

© NCERT not to be republished

invention of the Judiciary. How did it invent such a non-existent thing? And how is it that all other institutions have accepted this during the past four decades?

Therein lies the distinction between letter and spirit. The Court came to the conclusion that in reading a text or document, we must respect the intent behind that document. A mere text of the law is less important than the social circumstances and aspirations that have produced that law or document. The Court was looking at the basic structure as something without which the Constitution cannot be imagined at all. This is an instance of trying to balance the letter and the spirit of the Constitution.



Of course, if there are no rights and no elections, the Constitution won't make much sense. And if there is no well being, elections and rights won't make sense. Is this how we understand the 'spirit' of our Constitution?

Maturity of the Political Leadership

Our discussion of the role of Judiciary, in the paragraph above, brings out one more fact. In the background of the fierce controversy that raged between 1967 and 1973, Parliament and the Executive also realised that a balanced and long term view was necessary. After the Supreme Court gave the ruling in the Kesavananda case some attempts were made to ask the Court to reconsider its ruling. When these failed, the 42nd amendment was made and parliamentary supremacy was asserted. But the Court again repeated its earlier stand in the Minerva Mills case (1980). Therefore, even four decades after the ruling in the Kesavananda case, this ruling has dominated our interpretation of the Constitution. Political parties, political leaders, the government, and Parliament, accepted the idea of inviolable basic structure. Even when there was talk about 'review' of the Constitution, that exercise could not cross the limits set by the theory of the basic structure.

When the Constitution was made, leaders and people of our country shared a common vision of India. In Nehru's famous speech at the time of independence, this vision was described as a tryst with destiny. In the Constituent Assembly also, all the leaders mentioned this vision: dignity and freedom of the individual, social and



Let us not ignore that there are many instances of political immaturity as well. Does one have to list these?

Even within the Constituent Assembly, there were some members who felt that this Constitution was not suited to the Indian situation:

"The ideals on which this draft constitution is framed have no manifest relation to the fundamental spirit of India. ...this Constitution ...would not prove suitable and would break down soon after being brought into operation."

Lakshminarayan Sahu, CAD, Vol. XI, p. 613, 17 November 1949

economic equality, well-being of all people, unity based on national integrity. This vision has not disappeared. People and leaders alike hold to the vision and hope to realize it. Therefore, the Constitution, based on this vision, has remained an object of respect and authority even after half a century. The basic values governing our public imagination remain intact.

Conclusion

There can still be debates about what constitutes basic structure. There is nothing wrong in such debates. We must remember that politics in a democracy is necessarily full of debates and differences. That is a sign of diversity, liveliness and openness. Democracy welcomes debates. At the same time, our political parties and leadership have shown maturity in setting limits to these debates. Because, politics is also about compromises and give-and-take. Extreme positions may be theoretically very correct and ideologically very attractive, but politics demands that everyone is prepared to moderate their extreme views, sharp positions and reach a common minimum ground. Only then democratic politics becomes possible. Politicians and the people of India have understood and practised these skills. That has made the experience of working of the democratic Constitution quite successful. Among the different organs of the government, there will always be competition over which one is more important than the others. They will also always fight over what constitutes

© NCERT not to be republished

is harmless. But when these practices are challenged or threatened, neglect of the underlying principles can be harmful. In short, to get a handle on current constitutional practice, to grasp their value and meaning, we may have no option but to go back in time to the Constituent Assembly debates and perhaps even further back in time to the colonial era. Therefore, we need to remember and keep revisiting the political philosophy underlying our Constitution.



This is tough. Why couldn't they plainly tell us what the philosophy of this Constitution is? How can ordinary citizens understand the philosophy if it is hidden like this?

What is the Political Philosophy of our Constitution?

It is hard to describe this philosophy in one word. It resists any single label because it is liberal, democratic, egalitarian, secular, and federal, open to community values, sensitive to the needs of religious and linguistic minorities as well as historically disadvantaged groups, and committed to building a common national identity.

Read a Cartoon



Shankar. Copyright: Children's Book Trust.

26 January 1950

While all ideas unfold on this playfield, democracy is the 'Umpire'.

In short, it is committed to freedom, equality, social justice, and some form of national unity. But underneath all this, there is a clear emphasis on peaceful and democratic measures for putting this philosophy into practice.

Individual freedom

The first point to note about the Constitution is its commitment to individual freedom. This commitment did not emerge miraculously out of calm deliberations around a table. Rather, it was the product of continuous intellectual and political activity of well over a century. As early as the beginning of the nineteenth century, Rammohan Roy protested against curtailment of the freedom of the press by the British colonial state. Roy argued that a state responsive to the needs of individuals must provide them the means by which their needs are communicated. Therefore, the state must permit unlimited liberty of publication. Likewise, Indians continued to demand a free press throughout the British rule.

It is not surprising therefore that freedom of expression is an integral part of the Indian Constitution. So is the freedom from arbitrary arrest. After all, the infamous Rowlatt Act, which the national movement opposed so vehemently, sought to deny this basic freedom. These and other individual freedoms such as freedom of conscience are part of the liberal ideology. On this basis, we can say that the Indian Constitution has a pretty strong liberal character. In the chapter on fundamental rights we have already seen how the Constitution values individual freedom. It might be recalled that for over forty years before the adoption of the Constitution, every single resolution, scheme, bill and report of the Indian National Congress mentioned individual rights, not just in passing but as a non-negotiable value.

Social Justice

When we say that the Indian Constitution is liberal, we do not mean that it is liberal only in the classical western sense. In the book on Political Theory, you will learn more about the idea of liberalism. Classical liberalism always privileges rights of the individuals over demands of social justice and community values.

Reprint 2026-27

Check your progress

State which of the following rights are part of individual freedom:

- Freedom of expression
- Freedom of religion
- Cultural and educational rights of minorities
- Equal access to public places

The liberalism of the Indian Constitution differs from this version in two ways. First, it was always linked to social justice. The best example of this is the provision for reservations for Scheduled Castes and Scheduled Tribes in the Constitution. The makers of the Constitution believed that the mere granting of the right to equality was not enough to overcome age-old injustices suffered by these groups or to give real meaning to their right to vote. Special constitutional measures were required to advance their interests. Therefore the constitution makers provided a number of special measures to protect the interests of Scheduled Castes and Scheduled Tribes such as the reservation of seats in legislatures. The Constitution also made it possible for the government to reserve public sector jobs for these groups.

Indian liberalism has two streams. The first stream began with Rammohan Roy. He emphasised individual rights, particularly the rights of women. The second stream included thinkers like K.C. Sen, Justice Ranade and Swami Vivekananda. They introduced the spirit of social justice within orthodox Hinduism. For Vivekananda, such a reordering of Hindu society could not have been possible without liberal principles. — **K.M. Panikkar**, *In Defence of Liberalism*, Bombay, Asia Publishing House, 1962.



And while talking of social justice, let us not forget the directive principles.

Respect for diversity and minority rights

The Indian Constitution encourages equal respect between communities. This was not easy in our country, first because communities do not always have a relationship of equality; they tend to have hierarchical relationships with one another (as in the case of caste). Second, when these communities do see each other as equals, they also tend to become rivals (as in the case of religious communities). This was a huge challenge for the makers of the Constitution: how to make communities liberal in their approach and foster a sense of equal respect among them under existing conditions of hierarchy or intense rivalry?

It would have been very easy to resolve this problem by not recognising communities at all, as most western liberal constitutions do. But this would have been unworkable and undesirable in our country. This is not because Indians are attached to communities more than others. Individuals everywhere also belong to cultural communities and every such community has its own values, traditions, customs and language shared by its members. For example, individuals in France or Germany belong to a linguistic community and are deeply attached to it. What makes us different is that we have more openly acknowledged the value of communities. More importantly, India is a land of multiple cultural communities. Unlike Germany or France we have several linguistic and religious communities. It was important to ensure that no one community systematically dominates others. This made it mandatory for our Constitution to recognise community based rights.

One such right is the right of religious communities to establish and run their own educational institutions. Such institutions may receive money from the government. This provision shows that the Indian Constitution does not see religion merely as a 'private' matter concerning the individual.



I have always wondered who I am. I have so many 'identities' in my bag: I have my religious identity, I have my linguistic identity, I have ties with my parental town, and of course, I am a student also.

© NCERT not to be republished

Secularism

Secular states are widely seen as treating religion as only a private matter. That is to say, they refuse to give religion public or official recognition. Does this mean that the Indian Constitution is not secular? This does not follow. Though the term 'secular' was not initially mentioned, the Indian Constitution has always been secular. The mainstream, western conception, of secularism means mutual exclusion of state and religion in order to protect values such as individual freedom and citizenship rights of individuals.

Again, this is something that you will learn more about in Political Theory. The term 'mutual exclusion' means this: both religion and state must stay away from the internal affairs of one another. The state must not intervene in the domain of religion; religion likewise should not dictate state policy or influence the conduct of the state. In other words, mutual exclusion means that religion and state must be *strictly* separated.

What is the purpose behind *strict* separation? It is to safeguard the freedom of individuals. States which lend support to organised religions make them more powerful than they already are. When religious organisations begin to control the religious lives of individuals, when they start dictating how they should relate to God or how they should pray, individuals may have the option of turning to the modern state for protecting their religious freedom, but what help would a state offer them if it has already joined hands with these organisations? To protect religious freedom of individuals, therefore, state must not help religious organisations. But at the same time, state should not tell religious organisations how to manage their affairs. That too can thwart religious freedom. The state must, therefore, not hinder religious organisations either. In short, states should neither help nor hinder religions. Instead, they should keep themselves at an arm's length from them. This has been the prevalent western conception of secularism.



Have they started teaching us the Political Theory course?

Conditions in India were different and to respond to the challenge they posed, the makers of the Constitution had to work out an alternative conception of secularism. They departed from the western model in two ways and for two different reasons.

- *Rights of Religious Groups*

  First, as mentioned already, they recognised that inter-community equality was as necessary as equality between individuals. This was because a person's freedom and sense of self-respect was directly dependent upon the status of her community. If one community was dominated by another, then its members would also be significantly less free. If, on the other hand, their relations were equal, marked by an absence of domination, then its members would also walk about with dignity, self-respect and freedom. Thus, the Indian Constitution grants rights to all religious communities such as the right to establish and maintain their educational institutions. Freedom of religion in India means the freedom of religion of both individuals and communities.

- *State's Power of Intervention*

  Second, separation in India could not mean mutual exclusion. Why is it so? Because, religiously sanctioned customs such as untouchability deprived individuals of the most basic dignity and self-respect. Such customs were so deeply rooted and pervasive that without active state intervention, there was no hope of their dissolution. The state simply had to interfere in the affairs of religion. Such intervention was not always negative. The state could also help religious communities by giving aid to educational institutions run by them. Thus, the state may help or hinder religious communities depending on which mode of action promotes values such as freedom and equality. In India separation between religion and state did not mean their mutual exclusion but rather



I would like to know whether finally, the state can regulate matters related to religion or not. Otherwise, there can be no religious reform.

© NCERT not to be republished

principled distance, a rather complex idea that allows the state to be distant from all religions so that it can intervene or abstain from interference, depending upon which of these two would better promote liberty, equality and social justice.

We have hitherto mentioned three core features — these can also be seen as the achievements — of our Constitution.

- First, our Constitution reinforces and reinvents forms of liberal individualism. This is an important achievement because this is done in the backdrop of a society where community values are often indifferent or hostile to individual autonomy.
- Second, our Constitution upholds the principle of social justice without compromising on individual liberties. The constitutional commitment to caste-based affirmative action programme shows how much ahead India was compared to other nations. Can one forget that affirmative action programmes in the U.S. were begun after the 1964 Civil Rights Act, almost two decades after they were constitutionally entrenched in India?
- Third, against the background of inter-communal strife, the Constitution upholds its commitment to group rights (the right to the expression of cultural particularity). This indicates that the framers of the Constitution were more than willing to face the challenges of what more than four decades later has come to be known as multiculturalism.

Universal franchise

Two other core features may also be regarded as achievements. First, it is no mean achievement to commit oneself to universal franchise, specially when there is widespread belief that traditional hierarchies in India are congealed and more or less impossible to eliminate, and when the right to vote has only recently been extended to women and to the working class in stable, Western democracies.

Once the idea of a nation took root among the elite, the idea of democratic self-government followed. Thus, Indian nationalism always conceived of a political order based on the will of every single member of society. The idea of universal franchise lay securely within the heart of nationalism. As early as the Constitution of India Bill (1895), the first non-official attempt at drafting a constitution for India, the author declared that every citizen, i.e., anyone born in

"The Assembly has adopted the principle of adult franchise with an abundant faith in the common man and the ultimate success of democratic rule and in the full belief that the introduction of democratic government on the basis of adult suffrage will... promote the well-being..."



Alladi Krishnaswami Ayyar, CAD, Vol. XI, p. 835, 23 November 1949



It's certainly a matter of pride that the principle of 'one man one vote' was accepted almost uncontested. Isn't it true that women had to struggle for their right to vote in many other countries?

India, had a right to take part in the affairs of the country and be admitted to public office. The Motilal Nehru Report (1928) reaffirms this conception of citizenship, reiterating that every person of either sex who has attained the age of twenty-one is entitled to vote for the House of Representatives or Parliament. Thus from very early on, universal franchise was considered as the most important and legitimate instrument by which the will of the nation was to be properly expressed.

Federalism

Second, by introducing the article concerning North-East (Art. 371), the Indian Constitution anticipates the very important concept of asymmetric federalism. We have seen in the chapter on federalism that the Constitution has created a strong central government. But despite this unitary bias of the Indian Constitution, there are important constitutionally embedded differences between the legal status and prerogatives of different sub-units within the same federation. Unlike the constitutional symmetry of American federalism, Indian federalism has been constitutionally asymmetric. To meet the specific

© NCERT not to be republished

needs and requirements of some sub-units, it was always part of the original design to have a unique relationship with them or to give them special status.

Under Article 371A, the privilege of special status was also accorded to the North-Eastern State of Nagaland. This Article not only confers validity on pre-existing laws within Nagaland, but also protects local identity through restrictions on immigration. Many other States too, are beneficiaries of such special provisions. According to the Indian Constitution, then, there is nothing bad about this differential treatment.

Although the Constitution did not originally envisage this, India is now a multi-lingual federation. Each major linguistic group is politically recognised and all are treated as equals. Thus, the democratic and linguistic federalism of India has managed to combine claims to unity with claims to cultural recognition. A fairly robust political arena exists that allows for the play of multiple identities that complement one another.



I am really impressed! Who says our Constitution is based on imitation?. In every 'borrowed' aspect, we have put our own distinct imprint.

National identity

Thus, the Constitution constantly reinforces a common national identity. In the chapter on federalism, you have studied how India strives to retain regional identities along with the national identity. It is clear from what is mentioned above that this common national identity was not incompatible with distinct religious or linguistic identities. The Indian Constitution tried to balance these various identities. Yet, preference was given to common identity under certain conditions. This is clarified in the debate over separate electorates based on religious identity which the Constitution rejects. Separate electorates were rejected not because they fostered difference between religious communities as such or because they endangered a simple notion of national unity but because

© NCERT not to be republished

they endangered a healthy national life. Rather than forced unity, our Constitution sought to evolve true fraternity, a goal dear to the heart of Dr. Ambedkar. As Sardar Patel put it, the main objective was to evolve 'one community'.

"But in the long run, it would be in the interest of all to forget that there is anything like majority or minority in this country and that in India there is only one community..."



Sardar Patel, CAD, Vol. VIII, p. 272, 25 May 1949

Procedural Achievements

All these five core features are what might be called the substantive achievements of the Constitution. However, there were also some procedural achievements.

- First, the Indian Constitution reflects a faith in political deliberation. We know that many groups and interests were not adequately represented in the Constituent Assembly. But the debates in the Assembly amply show that the makers of the Constitution wanted to be as inclusive in their approach as possible. This open-ended approach indicates the willingness of people to modify their existing preferences, in short, to justify outcomes by reference not to self-interest but to reasons. It also shows a willingness to recognise creative value in difference and disagreement.
- Second, it reflects a spirit of compromise and accommodation. These words, compromise and accommodation, should not always be seen with disapproval. Not all compromises are bad.

© NCERT not to be republished

If something of value is traded off for mere self-interest, then we naturally have compromised in the bad sense. However, if one value is partially traded off for another value, especially in an open process of free deliberation among equals, then the compromise arrived in this manner can hardly be objected to. We may lament that we could not have everything but to secure a bit of all things important cannot be morally blameworthy. Besides, a commitment to the idea that decisions on the most important issues must be arrived at consensually rather than by majority vote is equally morally commendable.



I understand compromises in the design of institutions, but how can conflicting principles be accommodated?

Criticisms

The Indian Constitution can be subjected to many criticisms of which three may be briefly mentioned: first, that it is unwieldy; second, that it is unrepresentative and third, that it is alien to our conditions.

The criticism that it is unwieldy is based on the assumption that the entire constitution of a country must be found in one compact document. But this is not true even of countries such as the US which do have a compact constitution. The fact is that a country's constitution is to be identified with a compact document and with other written documents with constitutional status. Thus, it is possible to find important constitutional statements and practices outside one compact document. In the case of India, many such details, practices and statements are included in one single document and this has made that document somewhat large in size. Many countries for instance, do not have provisions for election commission or the civil service commission in the document known as constitution. But in India, many such matters are attended to by the Constitutional document itself.

A second criticism of the Constitution is that it is unrepresentative. Do you remember how the Constituent Assembly was formed? At that time, adult franchise was

not yet granted and most members came from the advanced sections of the society. Does this make our Constitution unrepresentative?

Here we must distinguish two components of representation, one that might be called voice and the other opinion. The voice component of representation is important. People must be recognised in their own language or voice, not in the language of the masters. If we look at the Indian Constitution from this dimension, it is indeed unrepresentative because members of the Constituent Assembly were chosen by a restricted franchise, not by universal suffrage. However, if we examine the other dimension, we may not find it altogether lacking in representativeness. The claim that almost every shade of opinion was represented in the Constituent Assembly may be a trifle exaggerated but may have something to it. If we read the debates that took place in the Constituent Assembly, we find that a vast range of issues and opinions were mentioned, members raised matters not only based on their individual social concerns but based on the perceived interests and concerns of various social sections as well.



Of course! Isn't it what we learnt in the first chapter? That there should be a valid reason for every section of society to go along with the Constitution?

Is it a coincidence that the central square of every other small town has a statue of Dr. Ambedkar with a copy of the Indian Constitution? Far from being a mere symbolic tribute to him, this expresses the feeling among Dalits that the Constitution reflects many of their aspirations.

A final criticism alleges that the Indian Constitution is entirely an alien document, borrowed article by article from western constitutions and sits uneasily with the cultural ethos of the Indian people. This criticism is often voiced by many. Even in the Constituent Assembly itself, there were some voices that echo this concern.

How far is this charge true?

It is true that the Indian Constitution is modern and partly western. Do you remember that in the first chapter we have listed the various sources from which our Constitution 'borrowed'? But in this chapter you have also

Reprint 2026-27

earns Rs 12,000. Can his standard of living be said to have risen 12 times during this period? By how much should his salary be raised so that he is as well off as before?

Case 2

You must be reading about the sensex in the newspapers. The sensex crossing 8000 points is, indeed, greeted with euphoria. When, sensex dipped 600 points recently, it eroded investors' wealth by Rs 1,53,690 crores. What exactly is sensex?

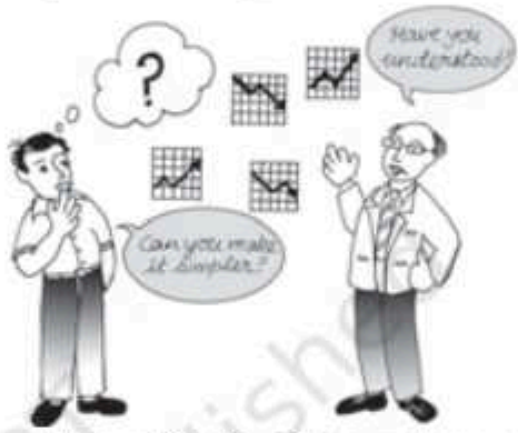
Case 3

The government says inflation rate will not accelerate due to the rise in the price of petroleum products. How does one measure inflation?

These are a sample of questions you confront in your daily life. A study of the index number helps in analysing these questions.

2. WHAT IS AN INDEX NUMBER

An index number is a statistical device for measuring changes in the magnitude of a group of related variables. It represents the general trend of diverging ratios, from which it is calculated. It is a measure of the average change in a group of related variables over two different situations. The comparison may be between like categories such as persons, schools, hospitals etc. An index number also measures changes in the value of the variables such as prices of specified list of commodities, volume of production in different sectors of an industry, production of various agricultural crops, cost of living etc.



Conventionally, index numbers are expressed in terms of percentage. Of the two periods, the period with which the comparison is to be made, is known as the base period. The value in the base period is given the index number 100. If you want to know how much the price has changed in 2005 from the level in 1990, then 1990 becomes the base. The index number of any period is in proportion with it. Thus an index number of 250 indicates that the value is two and half times that of the base period.

Price index numbers measure and permit comparison of the prices of certain goods. Quantity index numbers measure the changes in the physical volume of production, construction or employment. Though price index numbers are more widely used, a production index is also an important indicator of the level of the output in the economy.

© NCERT not to be republished

3. CONSTRUCTION OF AN INDEX NUMBER

In the following sections, the principles of constructing an index number will be illustrated through price index numbers.

Let us look at the following example:

Example 1

Calculation of simple aggregative price index

TABLE 7.1

Commodity	*Base period price (Rs)*	*Current period price (Rs)*	*Percentage change*
A	2	4	100
B	5	6	20
C	4	5	25
D	2	3	50

As you observe in this example, the percentage changes are different for every commodity. If the percentage changes were the same for all four items, a single measure would have been sufficient to describe the change. However, the percentage changes differ and reporting the percentage change for every item will be confusing. It happens when the number of commodities is large, which is common in any real market situation. *A price index represents these changes by a single numerical measure.*

There are two methods of constructing an index number. It can be computed by the *aggregative method* and by the *method of averaging relatives.*

The Aggregative Method

The formula for a *simple aggregative price index* is

$$P_{01} = \frac{\Sigma P_1}{\Sigma P_0} \times 100$$

Where P_1 and P_0 indicate the price of the commodity in the current period and base period respectively. Using the data from example 1, the *simple aggregative price index* is

$$P_{01} = \frac{4+6+5+3}{2+5+4+2} \times 100 = 138.5$$

Here, price is said to have risen by 38.5 per cent.

Do you know that such an index is of limited use? The reason is that the units of measurement of prices of various commodities are not the same. It is unweighted, because the relative importance of the items has not been properly reflected. The items are treated as having equal importance or weight. But what happens in reality? In reality the items purchased differ in order of importance. Food items occupy a large proportion of our expenditure. In that case an equal rise in the price of an item with large weight and that of an item with low weight will have different implications for the overall change in the price index.

The formula for a *weighted aggregative price index* is

$$P_{01} = \frac{\Sigma P_1 q_0}{\Sigma P_0 q_0} \times 100$$

An index number becomes a weighted index when the relative importance of items is taken care of.

Here weights are quantity weights. To construct a weighted aggregative index, a well-specified basket of commodities is taken and its worth each year is calculated. It thus measures the changing value of a fixed aggregate of goods. Since the total value changes with a fixed basket, the change is due to price change. Various methods of calculating a weighted aggregative index use different baskets with respect to time.



Example 2

Calculation of *weighted aggregative price index*

TABLE 7.2

Commodity	*Base period*		*Current period*	
	Price P_0	*Quantity* q_0	*Price* p_1	*Quantity* q_1
A	2	10	4	5
B	5	12	6	10
C	4	20	5	15
D	2	15	3	10

$$P_{01} = \frac{\Sigma P_1 q_0}{\Sigma P_0 q_0} \times 100$$

$$= \frac{4\times10 + 6\times12 + 5\times20 + 3\times15}{2\times10 + 5\times12 + 4\times20 + 2\times15} \times 100$$

$$= \frac{257}{190} \times 100 = 135.3$$

This method uses the base period quantities as weights. A weighted aggregative price index using base period quantities as weights, is also known as *Laspeyre's price index*. It provides an explanation to the question that if the expenditure on base period basket of commodities was Rs 100, how much should be the expenditure in the current period on the same basket of commodities? As you can see here, the value of base period quantities has risen by 35.3 per cent due to price rise. Using base period quantities as weights, the price is said to have risen by 35.3 percent.

Since the current period quantities differ from the base period quantities, the index number using current period weights gives a different value of the index number.

$$P_{01} = \frac{\Sigma P_1 q_1}{\Sigma P_0 q_1} \times 100$$

$$= \frac{4\times5 + 6\times10 + 5\times15 + 3\times10}{2\times5 + 5\times10 + 4\times15 + 2\times10} \times 100$$

$$= \frac{185}{140} \times 100 = 132.1$$

It uses the current period quantities as weights. A weighted aggregative price index using current period quantities as weights is known as *Paasche's price index*. It helps in answering the question that, if the

current period basket of commodities was consumed in the base period and if we were spending Rs 100 on it, how much should be the expenditure in current period on the same basket of commodities. Paasche's price index of 132.1 is interpreted as a price rise of 32.1 per cent. Using current period weights, the price is said to have risen by 32.1 per cent.

Method of Averaging relatives

When there is only one commodity, the price index is the ratio of the price of the commodity in the current period to that in the base period, usually expressed in percentage terms. The method of averaging relatives takes the average of these relatives when there are many commodities. The price index number using *price relatives* is defined as

$$P_{01} = \frac{1}{n} \Sigma \frac{p_1}{p_0} \times 100$$

where P_1 and P_0 indicate the price of the ith commodity in the current period and base period respectively. The ratio $(P_1/P_0) \times 100$ is also referred to as price relative of the commodity. *n* stands for the number of commodities. In the current exmple

$$P_{01} = \frac{1}{4}\left(\frac{4}{2} + \frac{6}{5} + \frac{5}{4} + \frac{3}{2}\right) \times 100 = 149$$

Thus, the prices of the commodities have risen by 49 per cent.

The *weighted index of price relatives* is the weighted arithmetic mean of price relatives defined as

$$P_{01} = \frac{\Sigma_{i=1}^{n} W_i \left(\frac{P_{1i}}{P_{0i}} \times 100\right)}{\Sigma_{i=1}^{n} W_i}$$

where W = Weight.

In a *weighted price relative index* weights may be determined by the proportion or percentage of expenditure on them in total expenditure during the base period. It can also refer to current period depending on the formula used. These are, essentially, the value shares of different commodities in the total expenditure. In general the base period weight is preferred to the current period weight. It is because calculating the weight every year is inconvenient. It also refers to the changing values of different baskets. They are strictly not comparable. *Example 3* shows the type of information one needs for calculating *weighted price index.*

Example 3

Calculation of *weighted price relatives index*

TABLE 7.3

Commodity	*Weight in %*	*Base year price price (in Rs.)*	*Current year (in Rs)*	*Price relative*
A	40	2	4	200
B	30	5	6	120
C	20	4	5	125
D	10	2	3	150

© NCERT not to be republished

The weighted price index is

$$P_{01} = \frac{\sum_{i=1}^{n} W_i \left(\frac{P_{1i}}{P_{0i}} \times 100 \right)}{\sum_{i=1}^{n} W_i}$$

$$= \frac{40 \times 200 + 30 \times 120 + 20 \times 125 + 10 \times 150}{100}$$

$$= 156$$

The *weighted price index* is 156. The price index has risen by 56 per cent. The values of the *unweighted price index* and the *weighted price index* differ, as they should. The higher rise in the weighted index is due to the doubling of the most important item A in *Example 3*.

Activity

- Interchange the current period values with the base period values, in the data given in *Example 2*. Calculate the price index using Laspeyre's, and Paasche's formula. What difference do you observe from the earlier illustration?

4. Some Important Index Numbers

Consumer price index

Consumer price index (CPI), also known as *the cost of living index*, measures the average change in retail prices. Consider the statement that the *CPI for industrial workers (2001=100) is 277 in December 2014*. What does this statement mean? It means that if the industrial worker was spending Rs 100 in 2001 for a typical basket of commodities, he needs Rs 277 in December 2014 to be able to buy an identical basket of commodities. It is not necessary that he/she buys the basket. What is important is whether he has the capability to buy it.

Example 4

Construction of *consumer price index* number.

$$CPI = \frac{\Sigma WR}{\Sigma W} = \frac{9786.85}{100} = 97.86$$

This exercise shows that the cost of living has declined by 2.14 per cent. What does an index larger than 100 indicate? It means a higher cost of living necessitating an upward adjustment in wages and salaries. The rise is equal to the amount, it exceeds 100. If the index is 150, 50 per cent upward adjustment is required. The salaries of the employees have to be raised by 50 per cent.

TABLE 7.4

Item	*Weight in % W*	*Base period price (Rs)*	*Current period price (Rs)*	*$R=P_1/P_0 \times 100$ (in%)*	*WR*
Food	35	150	145	96.67	3883.45
Fuel	10	25	23	92.00	920.00
Cloth	20	75	65	86.67	1733.40
Rent	15	30	30	100.00	1500.00
Misc.	20	40	45	112.50	2250.00
					9786.85

Consumer Price Index Number

Government agencies in India prepare a large number of consumer price index numbers. Some of them are as follows:

- Consumer Price Index Numbers for Industrial Workers with base 2001=100. Value of Index in May 2017 was 278.
- All-India Consumer Price Index Numbers for Agricultural Labourers with base 1986-87=100. Value of Index in May 2017 was 872.
- All-India Consumer Price Index Numbers for Rural Labourers with base 1986-87=100. Value of Index in May 2017 was 878.
- All-India Rural Consumer Index with base 2012 = 100. Value of Index in May 2017 was 133.3
- All-India Urban Consumer Price Index with base 2012 = 100. Value of Index in May 2017 was 129.3
  All-India Combined Consumer Price with base 2012 = 100. Value of Index in May 2017 was 131.4

In addition, these indices are available at the state level.

The detailed methods used for calculating each of these index numbers is different and it is not necessary to go into these details.

The Reserve Bank of India is using the All-India Combined Consumer Price Index as the main measure of how consumer prices are changing. Therefore, some details are necessary about this index number.

This index is now being prepared with base 2012 = 100 and many improvements have been made in accordance with international standards. The basket of items and weighing diagrams for the revised series has been prepared using the Modified Mixed Reference Period (MMRP) data of the Consumer Expenditure Survey (CES), 2011-12 of the 68th Round of National Sample Survey (NSS). The weights are as follows:

Major Groups	*Weight*
Food and beverages	45.86
Pan, tobacco and intoxicants	2.38
Clothing & footwear	6.53
Housing	10.07
Fuel & light	6.84
Misc. group	28.32
General	*100.00*

Source: *Economic Survey, 2014-15 Government of India.*

Data are provided on the rate of change per year of each of the sub-groups and main groups. So, we can find out from this data which prices are rising most of all and are, thereby, contributing to inflation.

The Consumer Food Price Index (CFPI) is the same as the Consumer Price Index for 'Food and Beverages' except that it does not include alcoholic beverages' and 'Prepared meals, snacks, sweets, etc'.

Wholesale Price Index

The Wholesale price index number indicates the change in the general price level. Unlike the CPI, it does not have any reference consumer category.

It does not include items pertaining to services like barber charges, repairing, etc.

What does the statement "WPI with 2004-05 as base is 253 in October, 2014" mean? It means that the general price level has risen by 153 per cent during this period.

The Wholesale Price Index is now being prepared with base 2011-12 = 100. The value of the index for May 2017 was 112.8. This index uses the prices that are prevailing at the wholesale level. Only the prices of goods are included. The main types of goods and their weights are as follows:

Major Groups	*Weight*
Primary Articles	22.62
Fuel and Power	13.15
Manufactured Products	64.23
All Commodities 'Headline Inflation'	100.00
'WPI Food Index'	*24.23*

Source: *Ministry of Statistics and Programme Implementation, 2016-17*

Usually the data on Wholesale Prices is available quickly. The 'All Commodities Inflation Rate' is often referred to as 'Headline Inflation'. Sometimes the focus is on food items which comprise 24.23% of the total weight. This Food Index is made up of Food Articles from the Primary Articles group and Food Products from the Manufactured Products group. Other economists like to focus on the wholesale prices in manufactured goods (other than food articles and also excluding fuel) and for this they study 'Core Inflation' which make up around 55% of the total weight of the wholesale price index.

Index of Industrial production

Unlike the Consumer Price Index or the Wholesale Price Index, this is an index which tries to measure quantities. With effect from April 2017, the base year has been fixed at 2011-12 = 100. The reason for the fast changes in the base year is that every year a large number of items either stop being manufactured or become inconsequential, while many other new items start getting manufactured.

While the price indices were essentially weighted averages of price relatives, the index of industrial production is a weighted arithmetic mean of quantity relatives with weights being allotted to various items in proportion to value added by manufacture in the base year by using Laspeyre's formula:

$$IIP_{01} = \frac{\sum_{i=1}^{n} q_{1i} W_i}{\sum_{i=1}^{n} W_i} \times 100$$

Where IIP_{01} is the index, q_{1i} is the quantity relative for year 1 with year 0 as base for good i, W_i is the weight allotted to the good i. There are n goods in the production index.

The index of Industrial Production is available at the level of Industrial Sectors and sub-sectors. The main branches are 'Mining', 'Manufacturing' and 'Electricity'. Sometimes the focus is on what are called "core" industries

namely coal, crude oil, natural gas, refinery products, fertilisers, steel, cement and electricity. The Eight Core Industries have a combined weight of 40.27 per cent in the IIP.

TABLE 7.5
Weightage Pattern of IIP (Industrial Production Sectors)

Sector	*Weight*
Mining	14.4
Manufacturing	77.6
Electricity	8.0
General Index	100.0

Source: *Ministry of Statistics and Programme Implementation, 2016-17*

The index of Industrial Production is also available according to the "use" of the product, that is, for example, "Primary Goods", "Consumer Durables" and so on.

TABLE 7.6
Weightage Pattern of IIP (Use-based Groups)

Group	*Weight*
Primary	34.1
Capital Goods	8.2
Intermediate Goods	17.2
Infrastructure/Construction Goods	12.3
Consumer Durables	12.8
Consumer Non-durables	15.3
General Index	100.0

Source: *Ministry of Statistics and Programme Implementation, 2016-17*

Human Development Index

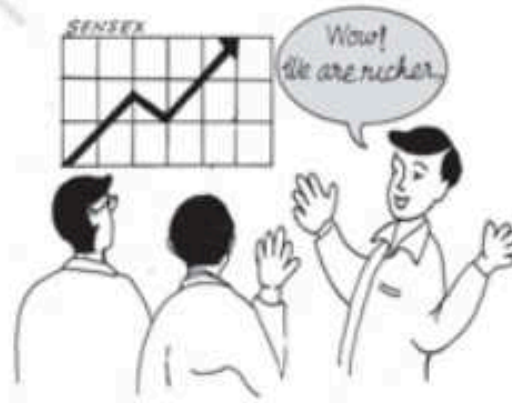
Another useful index widely used to know the development of a country is Human Development Index (HDI) about which you might have studied in Class X.

Sensex

Sensex is the short form of Bombay Stock Exchange Sensitive Index with 1978–79 as base. The value of the sensex is with reference to this period.



It is the benchmark index for the Indian stock market. It consists of 30 stocks which represent 13 sectors of the economy and the companies listed are leaders in their respective industries. If



the sensex rises, it indicates that the market is doing well and investors expect better earnings from companies. It also indicates a growing confidence of

© NCERT not to be republished

investors in the basic health of the economy.

5. Issues in the construction of an index number

You should keep certain important issues in mind, while constructing an index number.

- You need to be clear about the purpose of the index. Calculation of a volume index will be inappropriate, when one needs a value index.
- Besides this, the items are not equally important for different groups of consumers when a consumer price index is constructed. The rise in petrol price may not directly impact the living condition of the poor agricultural labourers. Thus the items to be included in any index have to be selected carefully to be as representative as possible. Only then you will get a meaningful picture of the change.
- Every index should have a base year. This base year should be as normal as possible. Years having extreme values should not be selected as base year. The period should also not belong to too far in the past. The comparison between 1993 and 2005 is much more meaningful than a comparison between 1960 and 2005. Many items in a 1960 typical consumption basket have disappeared at present. Therefore, the base year for any index number is routinely updated.
- Another issue is the choice of the formula, which depends on the nature of question to be studied. The only difference between the Laspeyre's index and Paasche's index is the weights used in these formulae.
- Besides, there are many sources of data with different degrees of reliability. Data of poor reliability will give misleading results. Hence, due care should be taken in the collection of data. If primary data are not being used, then the most reliable source of secondary data should be chosen.

Activity

- Collect data from the local vegetable market over a week for, at least 10 items. Try to construct the daily price index for the week. What problems do you encounter in applying both methods for the construction of a price index?

6. Index number in economics

Why do we need to use the index numbers? Wholesale price index number (WPI), consumer price index number (CPI) and industrial production index (IIP) are widely used in policy making.

- Consumer index number (CPI) or cost of living index numbers are helpful in wage negotiation, formulation of income policy, price policy, rent control, taxation and general economic policy formulation.
- The wholesale price index (WPI) is used to eliminate the effect of changes in prices on aggregates, such as national income, capital formation, etc.
- The WPI is widely used to measure the rate of inflation. Inflation is a general and continuing increase in prices. If inflation becomes sufficiently large,

money may lose its traditional function as a medium of exchange and as a unit of account. Its primary impact lies in lowering the value of money. The weekly inflation rate is given by

$\frac{X_t - X_{t-1}}{X_{t-1}} \times 100$ where X_t and X_{t-1} refer to the WPI for the tth and $(t-1)^{th}$ weeks.

- CPI are used in calculating the purchasing power of money and real wage:

(i) Purchasing power of money = 1/ Cost of living index

(ii) Real wage = (Money wage/Cost of living index) × 100

If the CPI (1982=100) is 526 in January 2005 the equivalent of a rupee in January, 2005 is given by

$\text{Rs}\ \frac{100}{526} = 0.19$. It means that it is worth 19 paise in 1982. If the money wage of the consumer is Rs 10,000, his real wage will be

$$\text{Rs}\ 10,000 \times \frac{100}{526} = \text{Rs}\ 1,901$$

It means Rs 1,901 in 1982 has the same purchasing power as Rs 10,000 in January, 2005. If he/she was getting Rs 3,000 in 1982, he/she is worse off due to the rise in price. To maintain the 1982 standard of living the salary should be raised to Rs 15,780 which is obtained by multiplying the base period salary by the factor 526/100.

- Index of industrial production gives us a quantitative figure about the change in production in the industrial sector.
- Agricultural production index provides us a ready reckoner of the performance of agricultural sector.
- Sensex is a useful guide for investors in the stock market. If the sensex is rising, investors are optimistic of the future performance of the economy. It is an appropriate time for investment.

Where can we get these index numbers?

Some of the widely used index numbers — WPI, CPI, Index Number of Yield of Principal Crops, Index of Industrial Production, Index of Foreign Trade — are available in *Economic Survey*.

Activity

- Check from the newspapers and construct a time series of sensex with 10 observations. What happens when the base of the consumer price index is shifted from 1982 to 2000?

7. Conclusion

Estimating index number enables you to calculate a single measure of change of a large number of items. Index numbers can be calculated for price, quantity, volume, etc.

It is also clear from the formulae that the index numbers need to be interpreted carefully. The items to be included and the choice of the base period are important. Index numbers are extremely important in policy making as is evident by their various uses.

© NCERT not to be republished

of people, villages are spread out over a relatively larger area. Villages are also distinguished from towns and cities by the larger share of agricultural activities in their economic profile. In other words, villages will have a significant proportion of its population engaged in agriculture linked occupations, much of what is produced there will be agricultural products, and most of its income will be from agriculture.

The distinction between a town and city is much more a matter of administrative definition. A town and city are basically the same sort of settlement, differentiated by size. An 'urban agglomeration' (a term used in Censuses and official reports) refers to a city along with its surrounding sub-urban areas and satellite settlements. A 'metropolitan area' includes more than one city, or a continuous urban settlement many times the size of a single city.

Given the directions in which modern societies have developed, the process of urbanisation has been experienced in most countries. This is the process by which a progressively larger and larger proportion of the country's population lives in urban rather than rural areas. Most developed countries are now overwhelmingly urban. Urbanisation is also the trend in developing countries: it can be faster or slower, but unless there are special reasons blocking it, the process does seem to occur in most contexts. According to United Nations report (2014), 54 per cent of the world's population lives in urban areas, a proportion that is expected to increase to 66 per cent by 2050 (United Nations, Department of Economc and Social Affairs, Population Division, 2014, World Population prospects). Indian society is also experiencing urbanisation: the percentage of the population living in urban areas has increased from a little less than 11 per cent in 1901 to a little more than17 per cent in 1951, soon after independence. The 2001 Census shows that almost 28 per cent of the population lives in urban areas. According to 2011 Census report, 37.7 per cent population of India lives in urban areas.

Social Order and Social Change in Rural Areas

Because of the objective conditions in villages being different, we can expect the nature of social order and social change to be different as well. Villages are small in size so they usually permit more personalised relationships; it is not unusual for members of a village to know all or most other members by sight. Moreover, the social structure in villages tends to follow a more traditional pattern: institutions like caste, religion, and other forms of customary or traditional social practice are stronger here. For these reasons, unless there are special circumstances that make for an exception, change is slower to arrive in villages than in towns.

There are also other reasons for this. A variety of factors ensure that the subordinate sections of society have much less scope for expressing themselves in rural areas than their counterparts in cities. The lack of

anonymity and distance in the village makes it difficult for people to dissent because they can be easily identified and 'taught a lesson' by the dominant sections. Moreover, the relative power of the dominant sections is much more because they control most avenues of employment, and most resources of all kinds. So the poor have to depend on the dominant sections since there are no alternative sources of employment or support. Given the small population, it is also very difficult to gather large numbers, particularly since efforts towards this cannot be hidden from the powerful and are very quickly suppressed. So, in short, if there is a strong power structure already in place in a village, it is very difficult to dislodge it. Change in the sense of shifts in power are thus slow and late to arrive in rural areas because the social order is stronger and more resilient.

Change of other sorts is also slow to come because villages are scattered and not as well connected to the rest of the world as cities and towns are. Of course, new modes of communication, particularly the telephone and the television have changed this. So the cultural 'lag' between villages and towns is now much shorter or non-existent. Communication links of other sorts (road, rail) have also generally improved over time so that few villages can really claim to be 'isolated' or 'remote', words often unthinkingly attached to villages in the past. This has also accelerated the pace of change somewhat.

For obvious reasons changes associated with agriculture or with agrarian social relations have a very major impact on rural societies. Thus, measures like land reform which alter the structure of land ownership have an immediate impact. In India, the first phase of land reforms after independence took away proprietary rights from absentee landlords and gave them to the groups that were actually managing the land and its cultivation in the village. Most of these groups belonged to intermediate castes, and though they were often not themselves the cultivators, they acquired rights over land. In combination with their number, this factor increased their social status and political power, because their votes mattered for winning elections. M.N. Srinivas has named these groups as the 'dominant castes'. In many regional contexts, the dominant castes became very powerful in economic terms and dominated the countryside and hence also electoral politics. In more recent times, these dominant castes are themselves facing opposition from the assertive uprisings of castes further below them, the lowest and the most backward castes. This has led to major social upheavals in many states like Andhra Pradesh, Bihar, Uttar Pradesh and Tamil Nadu.

In the same way, changes in the technological organisation of agriculture also has a large and immediate impact on rural society. The introduction of new labour saving machinery or new cropping patterns may alter the demand for labour and thus change the relative bargaining

strength of different social groups like landlords and labourers. Even if they don't directly affect labour demand, technological or economic changes can change the economic power of different groups and thus set in motion a chain of changes. Sudden fluctuations in agricultural prices, droughts or floods can cause havoc in rural society. The recent spate of farmer suicides in India is an example of this. On the other hand, large scale development programmes aimed at the rural poor can also have an enormous impact. A good example of this is the Mahatma Gandhi National Rural Employment Guarantee Act of 2005 (MGNREGA).

Activity 5

Find out more about the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA). What does it aim to do? Why is it considered such an important development programme? What problems does it face? What would be the likely consequences if it succeeds?

Social Order and Social Change in Urban Areas

It is well known that though the city itself is very old — even ancient societies had them — urbanism as a way of life for large segments of the population is a modern phenomenon. Before the modern era, trade, religion and warfare were some of the major factors that decided the location and importance of cities. Cities that were located on major trade routes, or had suitable harbours and ports had a natural advantage. So did cities that were well located from the point of view of military strategy. Finally, religious places attracted large numbers of pilgrims and thus supported an urban economy. In India too we have examples of such old cities, including the well known medieval trading towns of Tezpur on the Brahmaputra river in Assam or Kozhikode (formerly known as Calicut) on the Arabian Sea in northern Kerala. We also have many examples of temple towns and places of religious pilgrimage, such as Ajmer in Rajasthan, Varanasi (also known as Benaras or Kashi) in Uttar Pradesh, or Madurai in Tamil Nadu.

As sociologists have pointed out, city life and modernity go very well together; in fact, each may be considered an intimate expression of the other. Though it houses large and very dense populations, and though it has been known throughout history as the site for mass politics, the city is also the domain of the modern individual. In its combination of anonymity and the amenities and institutions that only large numbers can support, the city offers the individual boundless possibilities for fulfillment. Unlike the village, which discourages individuality and cannot offer much, the city nurtures the individual.

But while the many artists, writers, and scholars who have celebrated the city as the haven of the individual are not wrong, it is also true that freedom and opportunity are available only to some individuals. More accurately, only a socially and economically

privileged minority can have the luxury of a predominantly free and fulfilling life. Most people who live in cities have only limited and relative freedoms within larger constraints. These are the familiar economic and social constraints imposed by membership in social groups of various kinds, already known to you from the previous chapter. The city, too, fosters the development of group identities — based on factors like race, religion, ethnicity, caste, region, and of course class — which are all well represented in urban life. In fact, the concentration of large numbers in a relatively small space intensifies identities and makes them integral to strategies of survival, resistance and assertion.

Most of the important issues and problems of social order in towns and cities are related to the question of space. High population density places a great premium on space and creates very complex problems of logistics. It is the primary task of the urban social order to ensure the spatial viability of the city. This means the organisation and management of things like: housing and residential patterns; mass transit systems for transporting large numbers of workers to and fro for work; arranging for the coexistence of residential, public and industrial land-use zones; and finally all the public health, sanitation, policing, public safety and monitoring needs of urban governance. Each of these functions

A doctor checking a patient



© NCERT not to be republished

is a huge undertaking in itself and presents formidable challenges of planning, implementation and maintenance. What adds to the complexity is that all of these tasks have to be performed in a context where the divisions and tensions of class, ethnicity, religion, caste and so on are also present and active.

For example, the question of urban housing brings with it a whole host of problems. Shortage of housing for the poor leads to homelessness, and the phenomenon of 'street people' — those who live and survive on the streets and footpaths, under bridges and flyovers, abandoned buildings and other empty spaces. It is also the leading cause for the emergence of slums. Though official definitions vary, a slum is a congested, overcrowded neighbourhood with no proper civic facilities (sanitation, water supply, electricity and so on) and homes made of all kinds of building materials ranging from plastic sheets and cardboard to multi-storeyed concrete structures. Because of the absence of 'settled' property rights of the kind seen elsewhere, slums are the natural breeding ground for 'dadas' and strongmen who impose their authority on the people who live there. Control over slum territory becomes the natural stepping stone to other kinds of extra-legal activities, including criminal and real estate-related gangs.

A girl child looking after the sibling



A commercial centre in a city



Women at work in cotton field



The worldwide phenomenon of 'gated communities' is also found in Indian cities. This refers to the creation of affluent neighbourhoods that are separated from their surroundings by walls and gates, with controlled entry and exit. Most such communities also have their own parallel civic facilities, such as water and electricity supply, policing and security.

Activity 6

Have you come across such 'gated communities' in your town or city, or in one you have visited? Find out from your elders about such a community. When did the gates and fences come up? Was there any opposition, and if so by whom? What reasons might people have for wanting to live in such places? What effects do you think it has on urban society and on the neighbourhoods surrounding it?

Various kinds of transport in an urban area



Shopping in a city







Finally, housing patterns are linked to the economy of the city in crucial ways. The urban transport system is directly and severely affected by the location of residential areas relative to industrial and commercial workplaces. If these are far apart, as is often the case, an elaborate mass transit system must be created and maintained. Commuting becomes a way of life and an ever present source of possible disruption. The transport system has a direct impact on the 'quality of life' of working people in the city. Reliance on road transport and specially on private rather than public modes (i.e., cars rather than buses) creates problems of traffic congestion and vehicular pollution. As will be clear to you from the above discussion, the apparently simple issue of distribution of living space is actually a very complex and multi-dimensional aspect of urban society.

Daily long distance commuters can become an influential political constituency and sometimes develop elaborate sub-cultures. For example, the sub-urban trains of Mumbai — popularly known as 'locals' — have many informal associations of commuters. Collective on-train activities include singing *bhajans*, celebrating festivals, chopping vegetables, playing card and board games (including tournaments), or just general socialising.

The form and content of social change in urban areas is also best understood in relation to the central question of space. One very visible element of change is the ups and downs experienced by particular neighbourhoods and localities. Across the world, the city centre – or the core area of the original city – has had many changes of fortune. After being the power centre of the city in the 19th and early 20th century, the city centre went through a period of decline in the latter half of the 20th century. This was also the period of the growth of suburbs as the affluent classes deserted the inner city for the suburbs for a variety of reasons. City centres are experiencing a revival now in many major western cities as attempts to regenerate community life and the arts bear fruit. A related phenomenon is 'gentrification', which refers to the conversion of a previously lower class neighbourhood into a middle and upper class one. As real estate prices rise, it becomes more and more profitable for developers to try and effect such a conversion. At some point, the campaign becomes self-fulfilling as rental values increase and the locality acquires a critical minimum of prosperous businesses and residents. But sometimes the effort may fail and the neighbourhood goes back down the class scale and returns to its previous status.

Activity 7

Have you noticed any 'gentrification' or 'up-scaling' taking place in your neighbourhood? Do you know of such instances? Find out what the locality was like before this happened. In what ways has it changed? How have these changes affected different social groups and classes? Who benefits and who loses? Who decides about changes of this sort — is there voting, or some form of public discussion?

Changes in modes of mass transport may also bring about significant social change in cities. Affordable, efficient and safe public transport makes a huge difference to city life and can shape the social character of a city apart from influencing its economic fortunes. Many scholars have written on the difference between cities based on public transport like London or New York and cities that depend mainly on individualised car-based transport like Los Angeles. It remains to be seen, for example, whether the new Metro Rail in Delhi will significantly change social life in that city. But the main issue regarding social change in cities, specially in rapidly urbanising countries like India, is how the city will cope with constant increase in population as migrants keep streaming in to add to its natural growth.

GLOSSARY

Customs Duties, Tariffs: Taxes imposed on goods entering or leaving a country, which increase its price and make it less competitive relative to domestically produced goods.

Dominant Castes: Term attributed to M.N. Srinivas; refers to landowning intermediate castes that are numerically large and therefore enjoy political dominance in a given region.

Gated Communities: Urban localities (usually upper class or affluent) sealed off from its surroundings by fences, walls and gates, with controlled entry and exit.

Gentrification: The term used to describe the conversion of a low class (urban) neighbourhood into a middle or upper class neighbourhood.

© NCERT not to be republished

of complete nucleotide sequence of human genome during last decade has set in a new era of genomics. In the last section, the essentials of human genome sequencing and its consequences will also be discussed.

Let us begin our discussion by first understanding the structure of the most interesting molecule in the living system, that is, the DNA. In subsequent sections, we will understand that why it is the most abundant genetic material, and what its relationship is with RNA.

5.1 The DNA

DNA is a long polymer of deoxyribonucleotides. The length of DNA is usually defined as number of nucleotides (or a pair of nucleotide referred to as base pairs) present in it. This also is the characteristic of an organism. For example, a bacteriophage known as ϕ×174 has 5386 nucleotides, Bacteriophage lambda has 48502 base pairs (bp), *Escherichia coli* has 4.6×10^6 bp, and haploid content of human DNA is 3.3×10^9 bp. Let us discuss the structure of such a long polymer.

5.1.1 Structure of Polynucleotide Chain

Let us recapitulate the chemical structure of a polynucleotide chain (DNA or RNA). A nucleotide has three components – a nitrogenous base, a pentose sugar (ribose in case of RNA, and deoxyribose for DNA), and a phosphate group. There are two types of nitrogenous bases – Purines (Adenine and Guanine), and Pyrimidines (Cytosine, Uracil and Thymine). Cytosine is common for both DNA and RNA and Thymine is present in DNA. Uracil is present in RNA at the place of Thymine. A nitrogenous base is linked to the OH of 1′C pentose sugar through a N-glycosidic linkage to form a nucleoside, such as adenosine or deoxyadenosine, guanosine or deoxyguanosine, cytidine or deoxycytidine and uridine or deoxythymidine. When a phosphate group is linked to OH of 5′C of a nucleoside through phosphoester linkage, a corresponding nucleotide (or deoxynucleotide depending upon the type of sugar present) is formed. Two nucleotides are linked through 3′-5′ phosphodiester linkage to form a dinucleotide. More nucleotides can be joined in such a manner to form a polynucleotide chain. A polymer thus formed has at one end a free

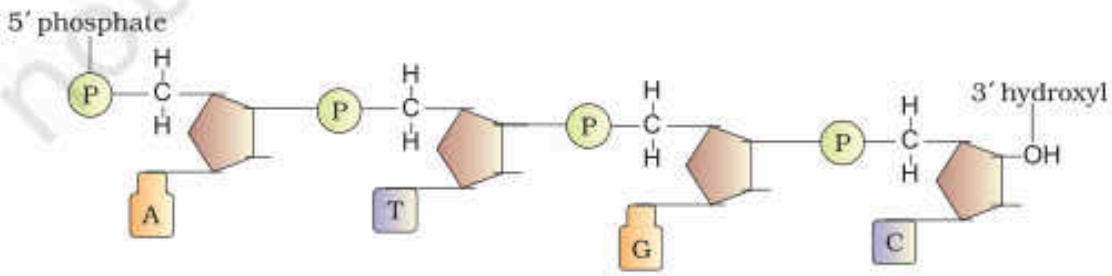


Figure 5.1 A Polynucleotide chain

Reprint 2026-27

phosphate moiety at 5′-end of sugar, which is referred to as 5'-end of polynucleotide chain. Similarly, at the other end of the polymer the sugar has a free OH of 3′C group which is referred to as 3′-end of the polynucleotide chain. The backbone of a polynucleotide chain is formed due to sugar and phosphates. The nitrogenous bases linked to sugar moiety project from the backbone (Figure 5.1).

In RNA, every nucleotide residue has an additional –OH group present at 2′-position in the ribose. Also, in RNA the uracil is found at the place of thymine (5-methyl uracil, another chemical name for thymine).

DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869. He named it as 'Nuclein'. However, due to technical limitation in isolating such a long polymer intact, the elucidation of structure of DNA remained elusive for a very long period of time. It was only in 1953 that James Watson and Francis Crick, based on the X-ray diffraction data produced by Maurice Wilkins and Rosalind Franklin, proposed a very simple but famous **Double Helix** model for the structure of DNA. One of the hallmarks of their proposition was base pairing between the two strands of polynucleotide chains. However, this proposition was also based on the observation of Erwin Chargaff that for a double stranded DNA, the ratios between **Adenine** and **Thymine** and **Guanine** and **Cytosine** are constant and equals one.

The base pairing confers a very unique property to the polynucleotide chains. They are said to be complementary to each other, and therefore if the sequence of bases in one strand is known then the sequence in other strand can be predicted. Also, if each strand from a DNA (let us call it as a parental DNA) acts as a template for synthesis of a new strand, the two double stranded DNA (let us call them as daughter DNA) thus, produced would be identical to the parental DNA molecule. Because of this, the genetic implications of the structure of DNA became very clear.

The salient features of the Double-helix structure of DNA are as follows:

(i) It is made of two polynucleotide chains, where the backbone is constituted by sugar-phosphate, and the bases project inside.

(ii) The two chains have anti-parallel polarity. It means, if one chain has the polarity 5′→3′, the other has 3′→5′.

(iii) The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. Similarly, Guanine is bonded with Cytosine with three H-bonds. As a result, always a purine comes opposite to a pyrimidine. This generates approximately uniform distance between the two strands of the helix (Figure 5.2).

(iv) The two chains are coiled in a right-handed fashion. The pitch of the helix is 3.4 nm (a nanometre is one billionth of a metre, that is 10^{-9} m) and there are roughly 10 bp in each

© NCERT not to be republished

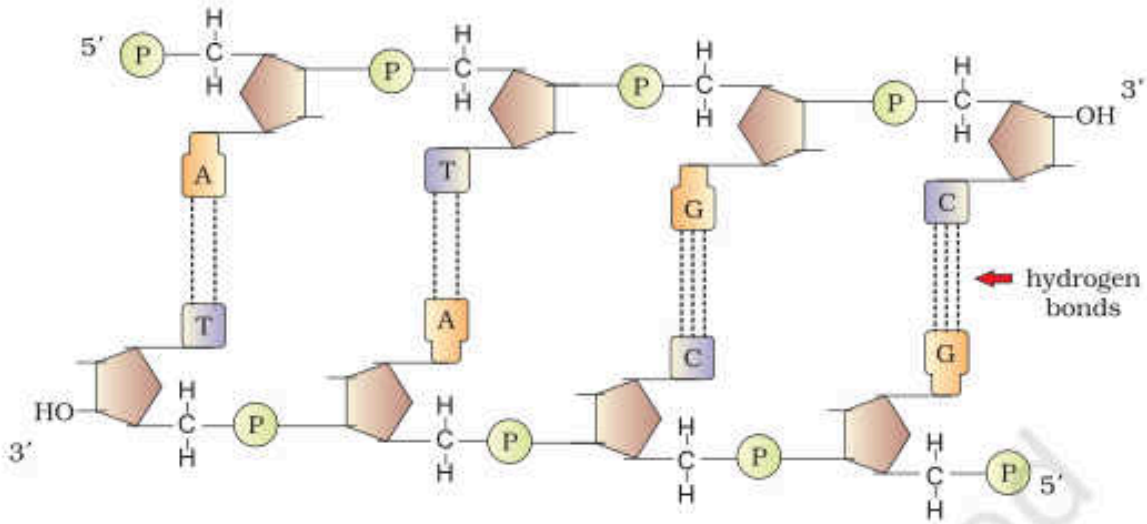


Figure 5.2 Double stranded polynucleotide chain

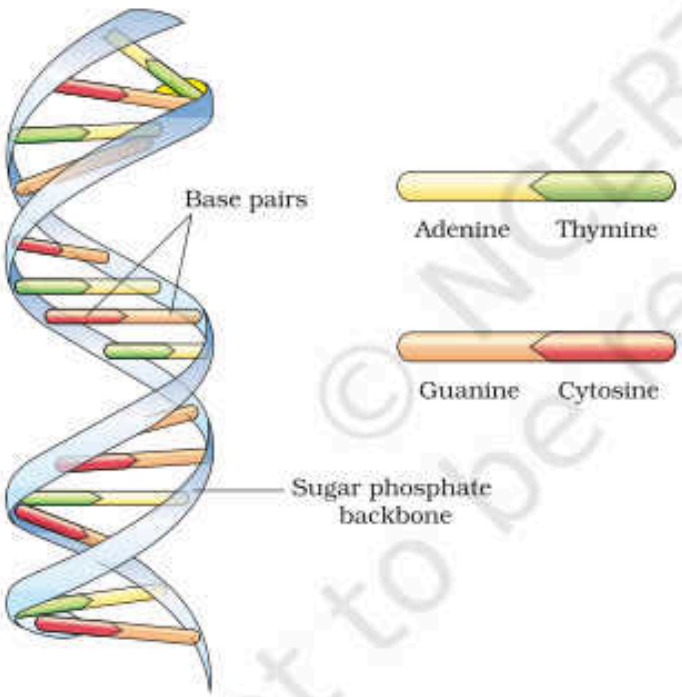


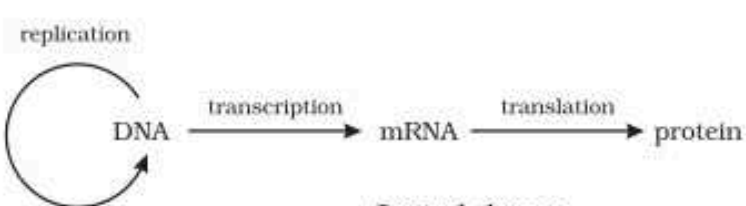
Figure 5.3 DNA double helix

turn. Consequently, the distance between a bp in a helix is approximately 0.34 nm.

(v) The plane of one base pair stacks over the other in double helix. This, in addition to H-bonds, confers stability of the helical structure (Figure 5.3).

Compare the structure of purines and pyrimidines. Can you find out why the distance between two polynucleotide chains in DNA remains almost constant?

The proposition of a double helix structure for DNA and its simplicity in explaining the genetic implication became revolutionary. Very soon, Francis Crick proposed the Central dogma in molecular biology, which states that the genetic information flows from DNA→RNA→Protein.



Central dogma

© NCERT not to be republished

In some viruses the flow of information is in reverse direction, that is, from RNA to DNA. *Can you suggest a simple name to the process?*

5.1.2 Packaging of DNA Helix

Taken the distance between two consecutive base pairs as 0.34 nm (0.34×10^{-9} m), if the length of DNA double helix in a typical mammalian cell is calculated (simply by multiplying the total number of bp with distance between two consecutive bp, that is, 6.6×10^{9} bp × 0.34×10^{-9} m/bp), it comes out to be approximately 2.2 metres. A length that is far greater than the dimension of a typical nucleus (approximately 10^{-6} m). How is such a long polymer packaged in a cell?

If the length of E. coli DNA is 1.36 mm, can you calculate the number of base pairs in E.coli?

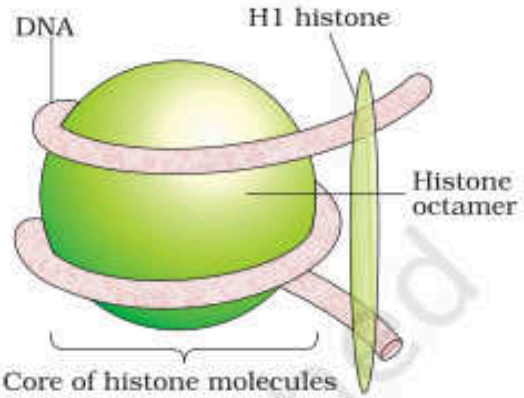


Figure 5.4a Nucleosome

In prokaryotes, such as, *E. coli*, though they do not have a defined nucleus, the DNA is not scattered throughout the cell. DNA (being negatively charged) is held with some proteins (that have positive charges) in a region termed as 'nucleoid'. The DNA in nucleoid is organised in large loops held by proteins.

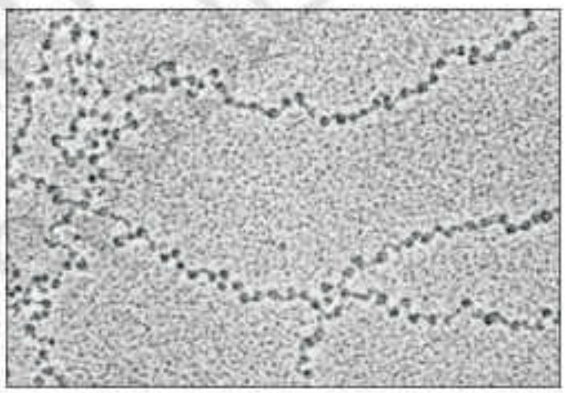


Figure 5.4b EM picture - 'Beads-on-String'

In eukaryotes, this organisation is much more complex. There is a set of positively charged, basic proteins called **histones**. A protein acquires charge depending upon the abundance of amino acids residues with charged side chains. Histones are rich in the basic amino acid residues lysine and arginine. Both the amino acid residues carry positive charges in their side chains. Histones are organised to form a unit of eight molecules called **histone octamer**. The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called **nucleosome** (Figure 5.4 a). A typical nucleosome contains 200 bp of DNA helix. Nucleosomes constitute the repeating unit of a structure in nucleus called **chromatin,** thread-like stained (coloured) bodies seen in nucleus. The nucleosomes in chromatin are seen as 'beads-on-string' structure when viewed under electron microscope (EM) (Figure 5.4 b).

Theoretically, how many such beads (nucleosomes) do you imagine are present in a mammalian cell?

The beads-on-string structure in chromatin is packaged to form chromatin fibers that are further coiled and condensed at metaphase stage of cell division to form chromosomes. The packaging of chromatin at higher level requires additional set of proteins that collectively are referred to as

The term **health** is very frequently used by everybody. *How do we define it*? Health does not simply mean 'absence of disease' or 'physical fitness'. It could be defined as a state of complete physical, mental and social well-being. When people are healthy, they are more efficient at work. This increases productivity and brings economic prosperity. Health also increases longevity of people and reduces infant and maternal mortality.

Balanced diet, personal hygiene and regular exercise are very important to maintain good health. Yoga has been practised since time immemorial to achieve physical and mental health. Awareness about diseases and their effect on different bodily functions, vaccination (immunisation) against infectious diseases, proper disposal of wastes, control of vectors and maintenance of hygiene in food and water resources are necessary for achieving good health.

When the functioning of one or more organs or systems of the body is adversely affected, characterised by appearance of various signs and symptoms, we say that we are not healthy, i.e., we have a **disease**. Diseases can be broadly grouped into **infectious** and **non-infectious**. Diseases which are easily transmitted from one person to another, are called **infectious diseases**. Infectious diseases are very common and every one of us suffers from these at sometime or other. Some of the infectious diseases like AIDS are fatal. Among non-infectious diseases, cancer is the major cause of death. Drug and alcohol abuse also affect our health adversely.

7.1 Common Diseases in Humans

A wide range of organisms belonging to bacteria, viruses, fungi, protozoans, helminths, etc., could cause diseases in man. Such disease-causing organisms are called **pathogens**. Most parasites are therefore pathogens as they cause harm to the host by living in (or on) them. The pathogens can enter our body by various means, multiply and interfere with normal vital activities, resulting in morphological and functional damage. Pathogens have to adapt to life within the environment of the host. For example, the pathogens that enter the gut must know a way of surviving in the stomach at low pH and resisting the various digestive enzymes. A few representative members from different groups of pathogenic organisms are discussed here alongwith the diseases caused by them. Preventive and control measures against these diseases in general, are also briefly described.

Salmonella typhi is a pathogenic bacterium which causes **typhoid** fever in human beings. These pathogens generally enter the small intestine through food and water contaminated with them and migrate to other organs through blood. Sustained high fever (39° to 40°C), weakness, stomach pain, constipation, headache and loss of appetite are some of the common symptoms of this disease. Intestinal perforation and death may occur in severe cases. Typhoid fever could be confirmed by

© NCERT not to be republished

Widal test : A classic case in medicine, that of Mary Mallon nicknamed *Typhoid Mary*, is worth mentioning here. She was a cook by profession and was a typhoid carrier who continued to spread typhoid for several years through the food she prepared.

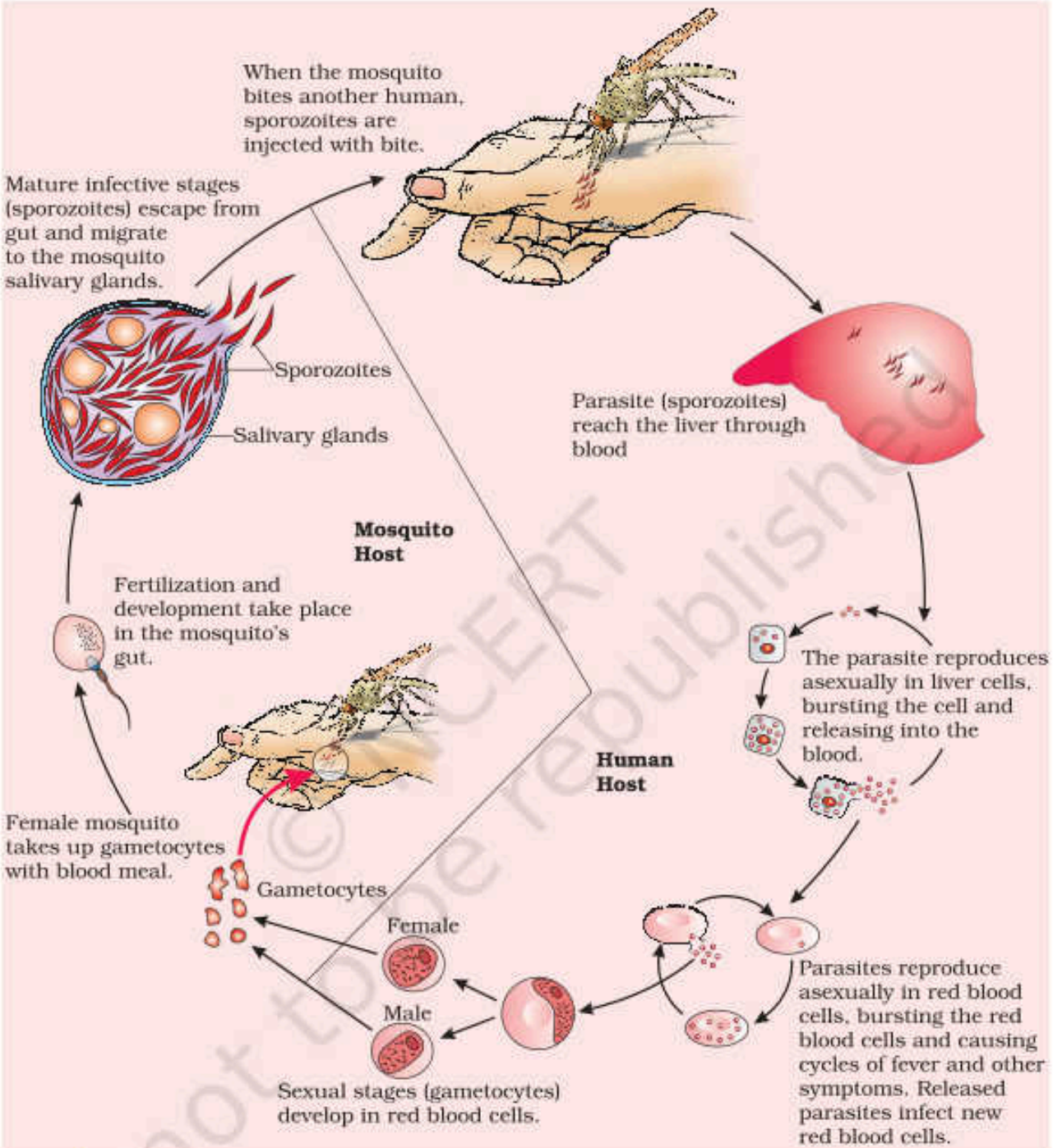
Bacteria like *Streptococcus pneumoniae* and *Haemophilus influenzae* are responsible for the disease **pneumonia** in humans which infects the alveoli (air filled sacs) of the lungs. As a result of the infection, the alveoli get filled with fluid leading to severe problems in respiration. The symptoms of pneumonia include fever, chills, cough and headache. In severe cases, the lips and finger nails may turn gray to bluish in colour. A healthy person acquires the infection by inhaling the droplets/aerosols released by an infected person or even by sharing glasses and utensils with an infected person. Dysentery, plague, diphtheria, etc., are some of the other bacterial diseases in man.

Many viruses also cause diseases in human beings. Rhino viruses represent one such group of viruses which cause one of the most infectious human ailments – the **common cold**. They infect the nose and respiratory passage but not the lungs. The common cold is characterised by nasal congestion and discharge, sore throat, hoarseness, cough, headache, tiredness, etc., which usually last for 3-7 days. Droplets resulting from cough or sneezes of an infected person are either inhaled directly or transmitted through contaminated objects such as pens, books, cups, doorknobs, computer keyboard or mouse, etc., and cause infection in a healthy person.

Some of the human diseases are caused by protozoans too. You might have heard about ***malaria***, a disease man has been fighting since many years. *Plasmodium*, a tiny protozoan is responsible for this disease. Different species of *Plasmodium* (*P. vivax*, *P. malaria* and *P. falciparum*) are responsible for different types of malaria. Of these, malignant malaria caused by *Plasmodium falciparum* is the most serious one and can even be fatal.

Let us take a glance at the life cycle of *Plasmodium* (Figure 7.1). *Plasmodium* enters the human body as sporozoites (infectious form) through the bite of infected female *Anopheles* mosquito. The parasites initially multiply within the liver cells and then attack the red blood cells (RBCs) resulting in their rupture. The rupture of RBCs is associated with release of a toxic substance, haemozoin, which is responsible for the chill and high fever recurring every three to four days. When a female *Anopheles* mosquito bites an infected person, these parasites enter the mosquito's body and undergo further development. The parasites multiply within them to form sporozoites that are stored in their salivary glands. When these mosquitoes bite a human, the sporozoites are introduced into his/her body, thereby initiating the events mentioned above. It is interesting to note that the malarial parasite requires two hosts – human and mosquitoes – to complete its life cycle (Figure 7.1); the female *Anopheles* mosquito is the vector (transmitting agent) too.

Reprint 2026-27



© NCERT not to be republished

Figure 7.1 Stages in the life cycle of *Plasmodium*

Entamoeba histolytica is a protozoan parasite in the large intestine of human which causes **amoebiasis (amoebic dysentery)**. Symptoms of this disease include constipation, abdominal pain and cramps, stools with excess mucous and blood clots. Houseflies act as mechanical carriers and serve to transmit the parasite from faeces of infected person to food

and food products, thereby contaminating them. Drinking water and food contaminated by the faecal matter are the main source of infection.

Ascaris, the common round worm and *Wuchereria*, the filarial worm, are some of the helminths which are known to be pathogenic to man. *Ascaris*, an intestinal parasite causes **ascariasis**. Symptoms of these disease include internal bleeding, muscular pain, fever, anemia and blockage of the intestinal passage. The eggs of the parasite are excreted along with the faeces of infected persons which contaminate soil, water, plants, etc. A healthy person acquires this infection through contaminated water, vegetables, fruits, etc.

Wuchereria (W. bancrofti and *W. malayi)*, the filarial worms cause a slowly developing chronic inflammation of the organs in which they live for many years, usually the lymphatic vessels of the lower limbs and the disease is called **elephantiasis** or **filariasis** (Figure 7.2). The genital organs are also often affected, resulting in gross deformities. The pathogens are transmitted to a healthy person through the bite by the female mosquito vectors.



Figure 7.2 Diagram showing inflammation in one of the lower limbs due to elephantiasis

Many fungi belonging to the genera *Microsporum*, *Trichophyton* and *Epidermophyton* are responsible for **ringworms** which is one of the most common infectious diseases in man. Appearance of dry, scaly lesions on various parts of the body such as skin, nails and scalp (Figure 7.3) are the main symptoms of the disease. These lesions are accompanied by intense itching. Heat and moisture help these fungi to grow, which makes them thrive in skin folds such as those in the groin or between the toes. Ringworms are generally acquired from soil or by using towels, clothes or even the comb of infected individuals.



Figure 7.3 Diagram showing ringworm affected area of the skin

Maintenance of personal and public hygiene is very important for prevention and control of many infectious diseases. Measures for personal hygiene include keeping the body clean; consumption of clean drinking water, food, vegetables, fruits, etc. Public hygiene includes proper disposal of waste and excreta; periodic cleaning and disinfection of water reservoirs, pools, cesspools and tanks and observing standard practices of hygiene in public catering. These measures are particularly essential where the infectious agents are transmitted through food and water such as typhoid, amoebiasis and ascariasis. In cases of air-borne diseases such as pneumonia and common cold, in addition to the above measures, close

© NCERT not to be republished

contact with the infected persons or their belongings should be avoided. For diseases such as malaria and filariasis that are transmitted through insect vectors, the most important measure is to control or eliminate the vectors and their breeding places. This can be achieved by avoiding stagnation of water in and around residential areas, regular cleaning of household coolers, use of mosquito nets, introducing fishes like *Gambusia* in ponds that feed on mosquito larvae, spraying of insecticides in ditches, drainage areas and swamps, etc. In addition, doors and windows should be provided with wire mesh to prevent the entry of mosquitoes. Such precautions have become more important especially in the light of recent widespread incidences of the vector-borne (*Aedes* mosquitoes) diseases like dengue and chikungunya in many parts of India.

The advancements made in biological science have armed us to effectively deal with many infectious diseases. The use of vaccines and immunisation programmes have enabled us to completely eradicate a deadly disease like smallpox. A large number of other infectious diseases like polio, diphtheria, pneumonia and tetanus have been controlled to a large extent by the use of vaccines. Biotechnology (about which you will read more in Chapter 10) is at the verge of making available newer and safer vaccines. Discovery of antibiotics and various other drugs has also enabled us to effectively treat infectious diseases.

7.2 Immunity

Everyday we are exposed to large number of infectious agents. However, only a few of these exposures result in disease. Why? This is due to the fact that the body is able to defend itself from most of these foreign agents. This overall ability of the host to fight the disease-causing organisms, conferred by the immune system is called **immunity**.

Immunity is of two types: (i) Innate immunity and (ii) Acquired immunity.

7.2.1 Innate Immunity

Innate immunity is non-specific type of defence, that is present at the time of birth. This is accomplished by providing different types of barriers to the entry of the foreign agents into our body. Innate immunity consist of four types of barriers. These are —

(i) ***Physical barriers*** : Skin on our body is the main barrier which prevents entry of the micro-organisms. Mucus coating of the epithelium lining the respiratory, gastrointestinal and urogenital tracts also help in trapping microbes entering our body.

(ii) ***Physiological barriers*** : Acid in the stomach, saliva in the mouth, tears from eyes–all prevent microbial growth.

(iii) ***Cellular barriers*** : Certain types of leukocytes (WBC) of our body like polymorpho-nuclear leukocytes (PMNL-neutrophils) and

monocytes and natural killer (type of lymphocytes) in the blood as well as macrophages in tissues can phagocytose and destroy microbes.

(iv) ***Cytokine barriers*** **:** Virus-infected cells secrete proteins called **interferons** which protect non-infected cells from further viral infection.

7.2.2 Acquired Immunity

Acquired immunity, on the other hand is pathogen specific. It is characterised by memory. This means when our body encounters a pathogen for the first time it produces a response called **primary response** which is of low intensity. Subsequent encounter with the same pathogen elicits a highly intensified secondary or anamnestic response. This is ascribed to the fact that our body appears to have memory of the first encounter.

The primary and secondary immune responses are carried out with the help of two special types of lymphocytes present in our blood, i.e., **B**-lymphocytes and **T**-lymphocytes.

The B-lymphocytes produce an army of proteins in response to pathogens into our blood to fight with them. These proteins are called antibodies. The T-cells themselves do not secrete antibodies but help B cells to produce them. Each antibody molecule has four peptide chains, two small called **light chains** and two longer called **heavy chains**. Hence, an antibody is represented as H_2L_2. Different types of antibodies are produced in our body. IgA, IgM, IgE, IgG are some of them. A cartoon of an antibody is given in Figure 7.4. Because these antibodies are found in the blood, the response is also called as **humoral immune response**. This is one of the two types of our acquired immune response – antibody mediated. The second type is called cell-mediated immune response or **cell-mediated immunity** (CMI). The T-lymphocytes mediate CMI. Very often, when some human organs like heart, eye, liver, kidney fail to function satisfactorily, transplantation is the only remedy to enable the patient to live a normal life. Then a search begins – to find a suitable donor. *Why is it that the organs cannot be taken from just anybody? What is it that the doctors check?* Grafts from

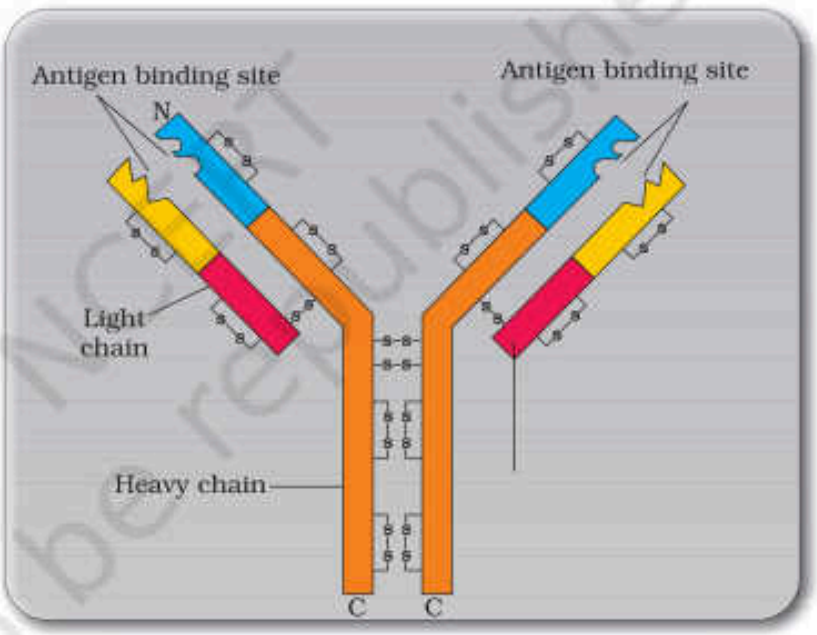


Figure 7.4 Structure of an antibody molecule

© NCERT not to be republished

just any source – an animal, another primate, or any human beings cannot be made since the grafts would be rejected sooner or later. Tissue matching, blood group matching are essential before undertaking any graft/transplant and even after this the patient has to take immuno–suppresants all his/her life. The body is able to differentiate 'self' and 'nonself' and the cell-mediated immune response is responsible for the graft rejection.

7.2.3 Active and Passive Immunity

When a host is exposed to antigens, which may be in the form of living or dead microbes or other proteins, antibodies are produced in the host body. This type of immunity is called **active immunity.** Active immunity is slow and takes time to give its full effective response. Injecting the microbes deliberately during immunisation or infectious organisms gaining access into body during natural infection induce active immunity. When ready-made antibodies are directly given to protect the body against foreign agents, it is called **passive immunity**. *Do you know why mother's milk is considered very essential for the new-born infant?* The yellowish fluid **colostrum** secreted by mother during the initial days of lactation has abundant antibodies (IgA) to protect the infant. The foetus also receives some antibodies from their mother, through the placenta during pregnancy. These are some examples of passive immunity.

7.2.4 Vaccination and Immunisation

The principle of immunisation or vaccination is based on the property of 'memory' of the immune system. In vaccination, a preparation of antigenic proteins of pathogen or inactivated/weakened pathogen (vaccine) are introduced into the body. The antibodies produced in the body against these antigens would neutralise the pathogenic agents during actual infection. The vaccines also generate memory – B and T-cells that recognise the pathogen quickly on subsequent exposure and overwhelm the invaders with a massive production of antibodies. If a person is infected with some deadly microbes to which quick immune response is required as in tetanus, we need to directly inject the preformed antibodies, or antitoxin (a preparation containing antibodies to the toxin). Even in cases of snakebites, the injection which is given to the patients, contain preformed antibodies against the snake venom. This type of immunisation is called **passive immunisation**.

Recombinant DNA technology has allowed the production of antigenic polypeptides of pathogen in bacteria or yeast. Vaccines produced using this approach allow large scale production and hence greater availability for immunisation, e.g., hepatitis B vaccine produced from yeast.

© NCERT not to be republished

7.2.5 Allergies

When you have gone to a new place and suddenly you started sneezing, wheezing for no explained reason, and when you went away, your symptoms dissappeared. Did this happen to you? Some of us are sensitive to some particles in the environment. The above-mentioned reaction could be because of allergy to pollen, mites, etc., which are different in different places.

The exaggerated response of the immune system to certain antigens present in the environment is called **allergy**. The substances to which such an immune response is produced are called allergens. The antibodies produced to these are of IgE type. Common examples of allergens are mites in dust, pollens, animal dander, etc. Symptoms of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing. Allergy is due to the release of chemicals like histamine and serotonin from the mast cells. For determining the cause of allergy, the patient is exposed to or injected with very small doses of possible allergens, and the reactions studied. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Somehow, modern-day life style has resulted in lowering of immunity and more sensitivity to allergens – more and more children in metro cities of India suffer from allergies and asthma due to sensitivity to the environment. This could be because of the protected environment provided early in life.

7.2.6 Auto Immunity

Memory-based acquired immunity evolved in higher vertebrates based on the ability to differentiate foreign organisms (e.g., pathogens) from self-cells. While we still do not understand the basis of this, two corollaries of this ability have to be understood. One, higher vertebrates can distinguish foreign molecules as well as foreign organisms. Most of the experimental immunology deals with this aspect. Two, sometimes, due to genetic and other unknown reasons, the body attacks self-cells. This results in damage to the body and is called **auto-immune** disease. Rheumatoid arthritis which affects many people in our society is an auto-immune disease.

7.2.7 Immune System in the Body

The human immune system consists of lymphoid organs, tissues, cells and soluble molecules like antibodies. As you have read, immune system is unique in the sense that it recognises foreign antigens, responds to these and remembers them. The immune system also plays an important role in allergic reactions, auto-immune diseases and organ transplantation.

Lymphoid organs: These are the organs where origin and/or maturation and proliferation of lymphocytes occur. The primary lymphoid organs are **bone marrow** and **thymus** where immature lymphocytes differentiate

© NCERT not to be republished

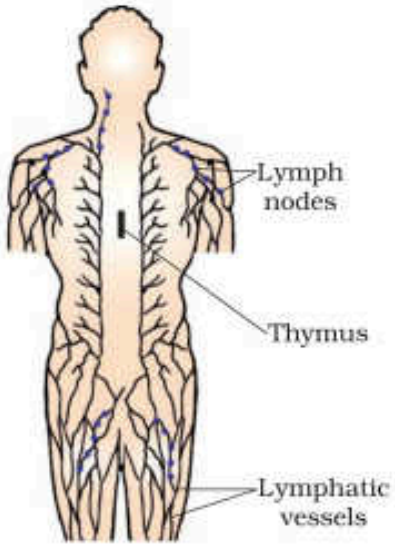


Figure 7.5 Diagrammatic representation of Lymph nodes

into antigen-sensitive lymphocytes. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen, which then proliferate to become effector cells. The location of various lymphoid organs in the human body is shown in Figure 7.5.

The bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced. The thymus is a lobed organ located near the heart and beneath the breastbone. The thymus is quite large at the time of birth but keeps reducing in size with age and by the time puberty is attained it reduces to a very small size. Both bone-marrow and thymus provide micro-environments for the development and maturation of T-lymphocytes. The spleen is a large bean-shaped organ. It mainly contains lymphocytes and phagocytes. It acts as a filter of the blood by trapping blood-borne micro-organisms. Spleen also has a large reservoir of erythrocytes. The lymph nodes are small solid structures located at different points along the lymphatic system. Lymph nodes serve to trap the micro-organisms or other antigens, which happen to get into the lymph and tissue fluid. Antigens trapped in the lymph nodes are responsible for the activation of lymphocytes present there and cause the immune response.

There is lymphoid tissue also located within the lining of the major tracts (respiratory, digestive and urogenital tracts) called **mucosa-associated lymphoid tissue** (MALT). It constitutes about 50 per cent of the lymphoid tissue in human body.

7.3 AIDS

The word AIDS stands for **Acquired Immuno Deficiency Syndrome**. This means deficiency of immune system, acquired during the lifetime of an individual indicating that it is not a congenital disease. 'Syndrome' means a group of symptoms. AIDS was first reported in 1981 and in the last twenty-five years or so, it has spread all over the world killing more than 25 million persons.

AIDS is caused by the Human Immuno deficiency Virus (HIV), a member of a group of viruses called **retrovirus**, which have an envelope enclosing the RNA genome (Figure 7.6). Transmission of HIV-infection generally occurs by (a) sexual contact with infected person, (b) by transfusion of contaminated blood and blood products, (c) by sharing infected needles as in the case of intravenous drug abusers and (d) from infected mother to her child through placenta. So, people who are at high risk of getting this infection includes - individuals who have multiple

Reprint 2026-27

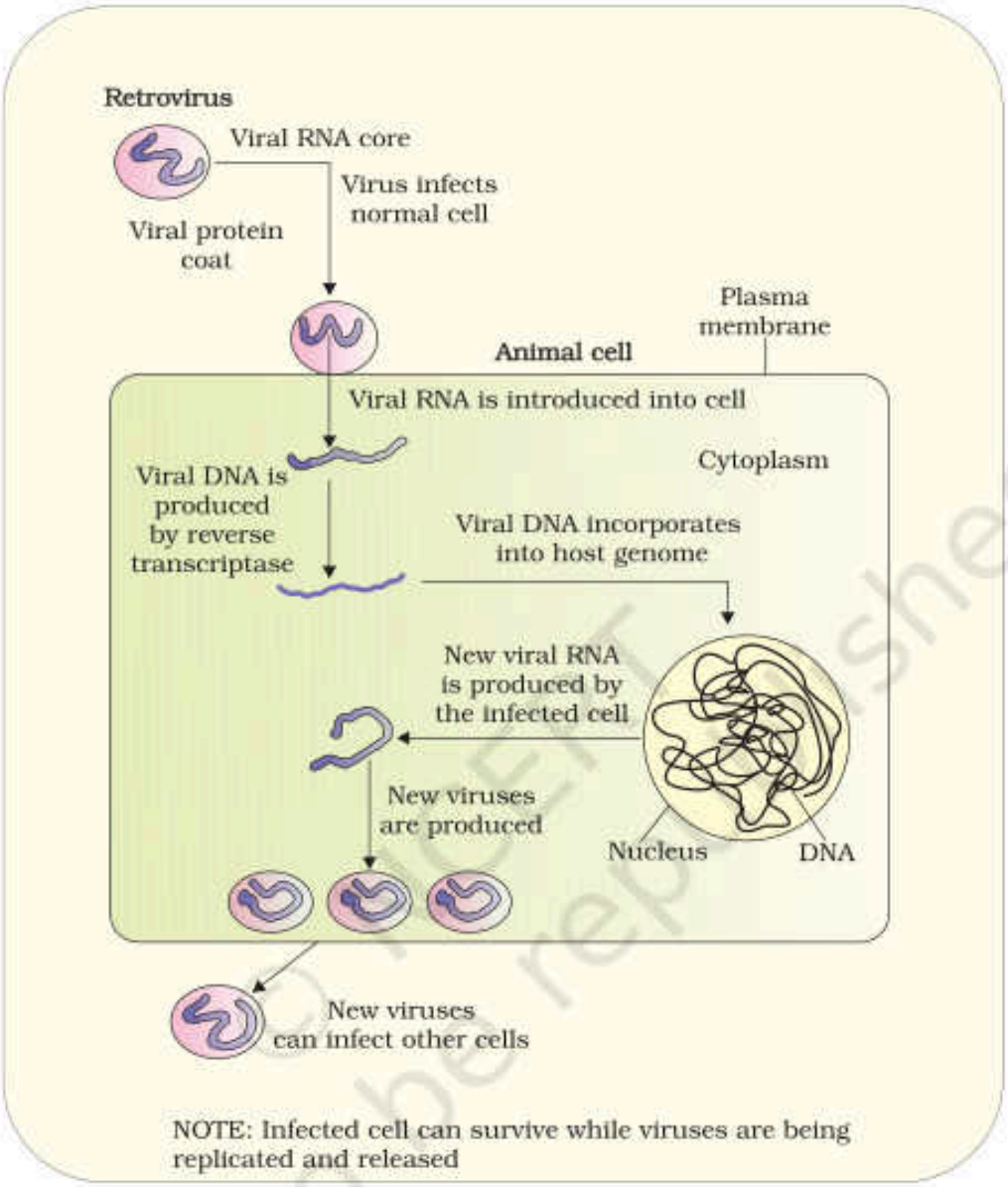


Figure 7.6 Replication of retrovirus

sexual partners, drug addicts who take drugs intravenously, individuals who require repeated blood transfusions and children born to an HIV infected mother. *Do you know–when do people need repeated blood transfusion? Find out and make a list of such conditions.* It is important to note that HIV/AIDS is not spread by mere touch or physical contact; it spreads only through body fluids. It is, hence, imperative, for the physical and psychological well-being, that the HIV/AIDS infected persons are not isolated from family and society. There is always a time-lag between the infection and appearance of AIDS symptoms. This period may vary from a few months to many years (usually 5-10 years).

© NCERT not to be republished

After getting into the body of the person, the virus enters into macrophages where RNA genome of the virus replicates to form viral DNA with the help of the enzyme reverse transcriptase. This viral DNA gets incorporated into host cell's DNA and directs the infected cells to produce virus particles (Figure 7.6). The macrophages continue to produce virus and in this way acts like a HIV factory. Simultaneously, HIV enters into helper T-lymphocytes (T_H), replicates and produce progeny viruses. The progeny viruses released in the blood attack other helper T-lymphocytes. This is repeated leading to a progressive decrease in the number of helper T-lymphocytes in the body of the infected person. During this period, the person suffers from bouts of fever, diarrhoea and weight loss. Due to decrease in the number of helper T lymphocytes, the person starts suffering from infections that could have been otherwise overcome such as those due to bacteria especially *Mycobacterium*, viruses, fungi and even parasites like *Toxoplasma*. The patient becomes so immuno-deficient that he/she is unable to protect himself/herself against these infections. A widely used diagnostic test for AIDS is **enzyme linked immuno-sorbent assay** (ELISA). Treatment of AIDS with anti-retroviral drugs is only partially effective. They can only prolong the life of the patient but cannot prevent death, which is inevitable.

Prevention of AIDS : As AIDS has no cure, prevention is the best option. Moreover, HIV infection, more often, spreads due to conscious behaviour patterns and is not something that happens inadvertently, like pneumonia or typhoid. Of course, infection in blood transfusion patients, new-borns (from mother) etc., may take place due to poor monitoring. The only excuse may be ignorance and it has been rightly said – "don't die of ignorance". In our country the National AIDS Control Organisation (NACO) and other non-governmental organisation (NGOs) are doing a lot to educate people about AIDS. WHO has started a number of programmes to prevent the spreading of HIV infection. Making blood (from blood banks) safe from HIV, ensuring the use of only disposable needles and syringes in public and private hospitals and clinics, free distribution of condoms, controlling drug abuse, advocating safe sex and promoting regular check-ups for HIV in susceptible populations, are some such steps taken up.

Infection with HIV or having AIDS is something that should not be hidden – since then, the infection may spread to many more people. HIV/AIDS-infected people need help and sympathy instead of being shunned by society. Unless society recognises it as a problem to be dealt with in a collective manner – the chances of wider spread of the disease increase manifold. It is a malady that can only be tackled, by the society and medical fraternity acting together, to prevent the spread of the disease.

7.4 Cancer

Cancer is one of the most dreaded diseases of human beings and is a major cause of death all over the globe. More than a million Indians suffer from

Reprint 2026-27

In chapter 7, you have read that microbes cause a large number of diseases in human beings. They also cause diseases in animals and plants. But this should not make you think that all microbes are harmful; several microbes are useful to man in diverse ways. Some of the most important contributions of microbes to human welfare are discussed in this chapter.

8.1 Microbes in Household Products

You would be surprised to know that we use microbes or products derived from them everyday. A common example is the production of curd from milk. Micro-organisms such as *Lactobacillus* and others commonly called **lactic acid bacteria (LAB)** grow in milk and convert it to curd. During growth, the LAB produce acids that coagulate and partially digest the milk proteins. A small amount of curd added to the fresh milk as inoculum or starter contain millions of LAB, which at suitable temperatures multiply, thus converting milk to curd, which also improves its nutritional quality by increasing vitamin B_{12}. In our stomach too, the LAB play very beneficial role in checking disease-causing microbes.

The dough, which is used for making foods such as *dosa* and *idli* is also fermented by bacteria. The puffed-up appearance of dough is due to the production of CO_2 gas. *Can you tell which metabolic pathway is taking place resulting in the formation of CO_2? Where do you think the bacteria for these fermentations come from?* Similarly the dough, which is used for making bread, is fermented using baker's yeast (*Saccharomyces cerevisiae*). A number of traditional drinks and foods are also made by fermentation by the microbes. 'Toddy', a traditional drink of some parts of southern India is made by fermenting sap from palms. Microbes are also used to ferment fish, soyabean and bamboo-shoots to make foods. Cheese, is one of the oldest food items in which microbes were used. Different varieties of cheese are known by their characteristic texture, flavour and taste, the specificity coming from the microbes used. For example, the large holes in 'Swiss cheese' are due to production of a large amount of CO_2 by a bacterium named *Propionibacterium sharmanii*. The 'Roquefort cheese' are ripened by growing a specific fungi on them, which gives them a particular flavour.

8.2 Microbes in Industrial Products

Even in industry, microbes are used to synthesise a number of products valuable to human beings. Beverages and antibiotics are some examples. Production on an industrial scale, requires growing microbes in very large vessels called **fermentors** (Figure 8.4).

© NCERT not to be republished



Figure 8.4 Fermentors



Figure 8.5 Fermentation Plant

8.2.1 Fermented Beverages

Microbes especially yeasts have been used from time immemorial for the production of beverages like wine, beer, whisky, brandy or rum. For this purpose the same yeast *Saccharomyces cerevisiae* used for bread-making and commonly called brewer's yeast, is used for fermenting malted cereals and fruit juices, to produce ethanol. *Do you recollect the metabolic reactions, which result in the production of ethanol by yeast?* Depending on the type of the raw material used for fermentation and the type of processing (with or without distillation) different types of alcoholic drinks are obtained. Wine and beer are produced without distillation whereas whisky, brandy and rum are produced by distillation of the fermented broth. The photograph of a fermentation plant is shown in Figure 8.5.

8.2.2 Antibiotics

Antibiotics produced by microbes are regarded as one of the most significant discoveries of the twentieth century and have greatly contributed towards the welfare of the human society. *Anti* is a Greek word that means 'against', and *bio* means 'life', together they mean 'against life' (in the context of disease causing organisms); whereas with reference to human beings, they are 'pro life' and not against. Antibiotics are chemical substances, which are produced by some microbes and can kill or retard the growth of other (disease-causing) microbes.

You are familiar with the commonly used antibiotic Penicillin. Do you know that Penicillin was the first antibiotic to be discovered, and it was a chance discovery? Alexander Fleming while working on *Staphylococci* bacteria, once observed a mould growing in one of his unwashed culture plates around which *Staphylococci* could not grow. He found out that it was due to a chemical produced by the mould and he named it Penicillin after the mould *Penicillium notatum*. However, its full potential as an effective antibiotic was established much later by Ernest Chain and Howard Florey. This antibiotic was extensively used to treat American soldiers wounded in World War II. Fleming, Chain and Florey were awarded the Nobel Prize in 1945, for this discovery.

© NCERT not to be republished

After Penicillin, other antibiotics were also purified from other microbes. *Can you name some other antibiotics and find out their sources?* Antibiotics have greatly improved our capacity to treat deadly diseases such as plague, whooping cough (*kali khansi*), diphtheria (*gal ghotu*) and leprosy (*kusht rog*), which used to kill millions all over the globe. Today, we cannot imagine a world without antibiotics.

8.2.3 Chemicals, Enzymes and other Bioactive Molecules

Microbes are also used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes. Examples of acid producers are *Aspergillus niger* (a fungus) of citric acid, *Acetobacter aceti* (a bacterium) of acetic acid; *Clostridium butylicum* (a bacterium) of butyric acid and *Lactobacillus* (a bacterium) of lactic acid.

Yeast (*Saccharomyces cerevisiae)* is used for commercial production of ethanol. Microbes are also used for production of enzymes. Lipases are used in detergent formulations and are helpful in removing oily stains from the laundry. You must have noticed that bottled fruit juices bought from the market are clearer as compared to those made at home. This is because the bottled juices are clarified by the use of pectinases and proteases. Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.

Another bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*. Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

8.3 Microbes in Sewage Treatment

We know that large quantities of waste water are generated everyday in cities and towns. A major component of this waste water is human excreta. This municipal waste-water is also called sewage. It contains large amounts of organic matter and microbes. Many of which are pathogenic. Have you ever wondered where this huge quantity of sewage or urban waste water is disposed off daily? This cannot be discharged into natural water bodies like rivers and streams directly – you can understand why. Before disposal, hence, sewage is treated in sewage treatment plants (STPs) to make it less polluting. Treatment of waste water is done by the

© NCERT not to be republished



Figure 8.6 Secondary treatment

heterotrophic microbes naturally present in the sewage. This treatment is carried out in two stages:

Primary treatment : These treatment steps basically involve physical removal of particles – large and small – from the sewage through filtration and sedimentation. These are removed in stages; initially, floating debris is removed by sequential filtration. Then the grit (soil and small pebbles) are removed by sedimentation. All solids that settle form the **primary sludge**, and the supernatant forms the effluent. The effluent from the primary settling tank is taken for secondary treatment.

Secondary treatment or Biological treatment : The primary effluent is passed into large aeration tanks (Figure 8.6) where it is constantly agitated mechanically and air is pumped into it. This allows vigorous growth of useful aerobic microbes into **flocs** (masses of bacteria associated with fungal filaments to form mesh like structures). While growing, these microbes consume the major part of the organic matter in the effluent. This significantly reduces the **BOD (biochemical oxygen demand)** of the effluent. BOD refers to the amount of the oxygen that would be consumed if all the organic matter in one liter of water were oxidised by bacteria. The sewage water is treated till the BOD is reduced. The BOD test measures the rate of uptake of oxygen by micro-organisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water. The greater the BOD of waste water, more is its polluting potential.

Once the BOD of sewage or waste water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment. This sediment is called **activated sludge**. A small part of the activated sludge is pumped back into the aeration tank to serve as the inoculum. The remaining major part of the sludge is pumped into large tanks called **anaerobic sludge digesters**. Here, other kinds of bacteria, which grow anaerobically, digest the bacteria and the fungi in the sludge. During this digestion, bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide. These gases form **biogas** and can be used as source of energy as it is inflammable.

The effluent from the secondary treatment plant is generally released into natural water bodies like rivers and streams. An aerial view of such a plant is shown in Figure 8.7.

Reprint 2026-27

You can appreciate how microbes play a major role in treating millions of gallons of waste water everyday across the globe. This methodology has been practiced for more than a century now, in almost all parts of the world. Till date, no man-made technology has been able to rival the microbial treatment of sewage.



Figure 8.7 An aerial view of a sewage plant

You are aware that due to increasing urbanisation, sewage is being produced in much larger quantities than ever before. However the number of sewage treatment plants has not increased enough to treat such large quantities. So the untreated sewage is often discharged directly into rivers leading to their pollution and increase in water-borne diseases.

The Ministry of Environment and Forests has initiated **Ganga Action Plan** and **Yamuna Action Plan** to save these major rivers of our country from pollution. Under these plans, it is proposed to build a large number of sewage treatment plants so that only treated sewage may be discharged in the rivers. A visit to a sewage treatment plant situated in any place near you would be a very interesting and educating experience.

8.4 Microbes in Production of Biogas

Biogas is a mixture of gases (containing predominantly methane) produced by the microbial activity and which may be used as fuel. You have learnt that microbes produce different types of gaseous end-products during growth and metabolism. The type of the gas produced depends upon the microbes and the organic substrates they utilise. In the examples cited in relation to fermentation of dough, cheese making and production of beverages, the main gas produced was CO_2. However, certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO_2 and H_2. These bacteria are collectively called **methanogens**, and one such common bacterium is *Methanobacterium*. These bacteria are commonly found in the anaerobic sludge during sewage treatment. These bacteria are also present in the rumen (a part of stomach) of cattle. A lot of cellulosic material present in the food of cattle is also present in the rumen. In rumen, these bacteria help in the breakdown of cellulose and play an important role in the nutrition of cattle. *Do you think we, human beings, are able to digest the cellulose present in our foods?* Thus, the excreta (dung) of cattle, commonly called *gobar*, is rich in these bacteria. Dung can be used for generation of biogas, commonly called *gobar gas*.

The biogas plant consists of a concrete tank (10-15 feet deep) in which bio-wastes are collected and a slurry of dung is fed. A floating cover is

© NCERT not to be republished

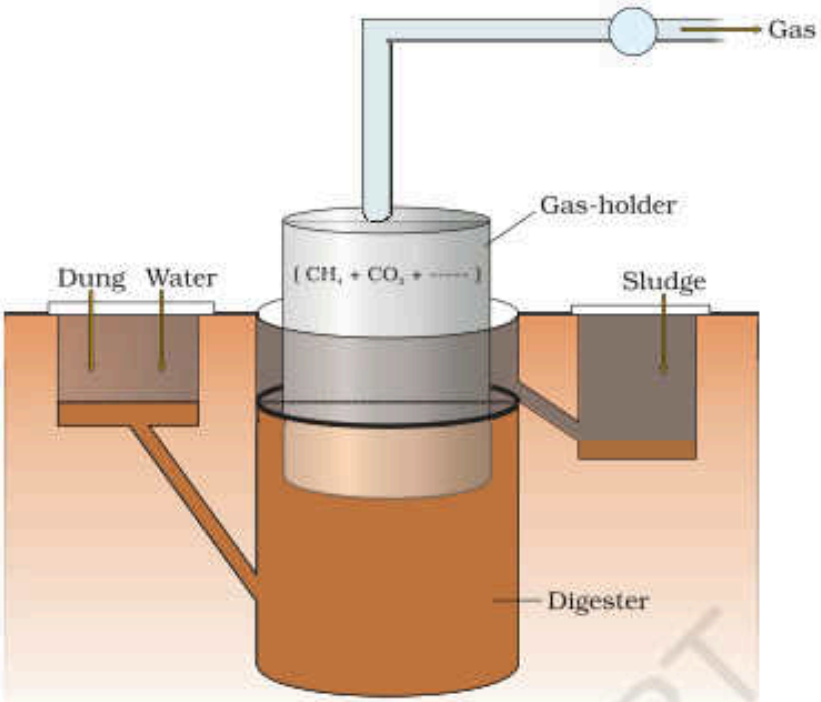


Figure 8.8 A typical biogas plant

placed over the slurry, which keeps on rising as the gas is produced in the tank due to the microbial activity. The biogas plant has an outlet, which is connected to a pipe to supply biogas to nearby houses. The spent slurry is removed through another outlet and may be used as fertiliser. Cattle dung is available in large quantities in rural areas where cattle are used for a variety of purposes. So biogas plants are more often built in rural areas. The biogas thus produced is used for cooking and lighting. The picture of a biogas plant is shown in Figure 8.8. The technology of biogas production was developed in India mainly due to the efforts of Indian Agricultural Research Institute (IARI) and Khadi and Village Industries Commission (KVIC). If your school is situated in a village or near a village, it would be very interesting to enquire if there are any biogas plants nearby. Visit the biogas plant and learn more about it from the people who are actually managing it.

8.5 Microbes as Biocontrol Agents

Biocontrol refers to the use of biological methods for controlling plant diseases and pests. In modern society, these problems have been tackled increasingly by the use of chemicals – by use of insecticides and pesticides. These chemicals are toxic and extremely harmful, to human beings and animals alike, and have been polluting our environment (soil, ground water), fruits, vegetables and crop plants. Our soil is also polluted through our use of weedicides to remove weeds.

Biological control of pests and diseases: In agriculture, there is a method of controlling pests that relies on natural predation rather than introduced chemicals. A key belief of the organic farmer is that biodiversity furthers health. The more variety a landscape has, the more sustainable it is. The organic farmer, therefore, works to create a system where the insects that are sometimes called pests are not eradicated, but instead are kept at manageable levels by a complex system of checks and balances within a living and vibrant ecosystem. Contrary to the 'conventional' farming practices which often use chemical methods to kill both useful

© NCERT not to be republished

and harmful life forms indiscriminately, this is a holistic approach that seeks to develop an understanding of the webs of interaction between the myriad of organisms that constitute the field fauna and flora. The organic farmer holds the view that the eradication of the creatures that are often described as pests is not only possible, but also undesirable, for without them the beneficial predatory and parasitic insects which depend upon them as food or hosts would not be able to survive. Thus, the use of biocontrol measures will greatly reduce our dependence on toxic chemicals and pesticides. An important part of the biological farming approach is to become familiar with the various life forms that inhabit the field, predators as well as pests, and also their life cycles, patterns of feeding and the habitats that they prefer. This will help develop appropriate means of biocontrol.

The very familiar beetle with red and black markings – the Ladybird, and Dragonflies are useful to get rid of aphids and mosquitoes, respectively. An example of microbial biocontrol agents that can be introduced in order to control butterfly caterpillars is the bacteria *Bacillus thuringiensis* (often written as *Bt*). These are available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants such as brassicas and fruit trees, where these are eaten by the insect larvae. In the gut of the larvae, the toxin is released and the larvae get killed. The bacterial disease will kill the caterpillars, but leave other insects unharmed. Because of the development of methods of genetic engineering in the last decade or so, the scientists have introduced *B. thuringiensis* toxin genes into plants. Such plants are resistant to attack by insect pests. **Bt-cotton** is one such example, which is being cultivated in some states of our country. You will learn more about this in chapter 10.

A biological control being developed for use in the treatment of plant disease is the fungus *Trichoderma*. *Trichoderma* species are free-living fungi that are very common in the root ecosystems. They are effective biocontrol agents of several plant pathogens.

Baculoviruses are pathogens that attack insects and other arthropods. The majority of baculoviruses used as biological control agents are in the genus *Nucleopolyhedrovirus*. These viruses are excellent candidates for species-specific, narrow spectrum insecticidal applications. They have been shown to have no negative impacts on plants, mammals, birds, fish or even on non-target insects. This is especially desirable when beneficial insects are being conserved to aid in an overall integrated pest management (IPM) programme, or when an ecologically sensitive area is being treated.

8.6 Microbes as Biofertilisers

With our present day life styles environmental pollution is a major cause of concern. The use of the chemical fertilisers to meet the ever-increasing

Reprint 2026-27

CHAPTER 9



BIOTECHNOLOGY : PRINCIPLES AND PROCESSES

9.1 *Principles of Biotechnology*

9.2 *Tools of Recombinant DNA Technology*

9.3 *Processes of Recombinant DNA Technology*

Biotechnology deals with techniques of using live organisms or enzymes from organisms to produce products and processes useful to humans. In this sense, making curd, bread or wine, which are all microbe-mediated processes, could also be thought as a form of biotechnology. However, it is used in a restricted sense today, to refer to such of those processes which use genetically modified organisms to achieve the same on a larger scale. Further, many other processes/techniques are also included under biotechnology. For example, *in vitro* fertilisation leading to a 'test-tube' baby, synthesising a gene and using it, developing a DNA vaccine or correcting a defective gene, are all part of biotechnology.

The European Federation of Biotechnology (EFB) has given a definition of biotechnology that encompasses both traditional view and modern molecular biotechnology. The definition given by EFB is as follows:

'The integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services'.

9.1 Principles of Biotechnology

Among many, the two core techniques that enabled birth of modern biotechnology are :

(i) ***Genetic engineering*** : Techniques to alter the chemistry of genetic material (DNA and RNA),

© NCERT not to be republished

to introduce these into host organisms and thus change the phenotype of the host organism.

(ii) ***Bioprocess engineering***: Maintenance of sterile (microbial contamination-free) ambience in chemical engineering processes to enable growth of only the desired microbe/eukaryotic cell in large quantities for the manufacture of biotechnological products like antibiotics, vaccines, enzymes, etc.

Let us now understand the conceptual development of the principles of genetic engineering.

You probably appreciate the advantages of sexual reproduction over asexual reproduction. The former provides opportunities for variations and formulation of unique combinations of genetic setup, some of which may be beneficial to the organism as well as the population. Asexual reproduction preserves the genetic information, while sexual reproduction permits variation. Traditional hybridisation procedures used in plant and animal breeding, very often lead to inclusion and multiplication of undesirable genes along with the desired genes. The techniques of genetic engineering which include creation of **recombinant DNA**, use of **gene cloning** and **gene transfer**, overcome this limitation and allows us to isolate and introduce only one or a set of desirable genes without introducing undesirable genes into the target organism.

Do you know the likely fate of a piece of DNA, which is somehow transferred into an alien organism? Most likely, this piece of DNA would not be able to multiply itself in the progeny cells of the organism. But, when it gets integrated into the genome of the recipient, it may multiply and be inherited along with the host DNA. This is because the alien piece of DNA has become part of a chromosome, which has the ability to replicate. In a chromosome there is a specific DNA sequence called the **origin of replication,** which is responsible for initiating replication. Therefore, for the multiplication of any alien piece of DNA in an organism it needs to be a part of a chromosome(s) which has a specific sequence known as 'origin of replication'. Thus, an alien DNA is linked with the origin of replication, so that, this alien piece of DNA can replicate and multiply itself in the host organism. This can also be called as **cloning** or making multiple identical copies of any template DNA.

Let us now focus on the first instance of the construction of an artificial recombinant DNA molecule. The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native **plasmid** (autonomously replicating circular extra-chromosomal DNA) of *Salmonella typhimurium*. Stanley Cohen and Herbert Boyer accomplished this in 1972 by isolating the antibiotic resistance gene by cutting out a piece of DNA from a plasmid which was responsible for conferring antibiotic resistance. The cutting of DNA at specific locations became possible with the discovery of the so-called

Reprint 2026-27

'molecular scissors'– **restriction enzymes**. The cut piece of DNA was then linked with the plasmid DNA. These plasmid DNA act as **vectors** to transfer the piece of DNA attached to it. You probably know that mosquito acts as an insect vector to transfer the malarial parasite into human body. In the same way, a plasmid can be used as vector to deliver an alien piece of DNA into the host organism. The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme DNA ligase, which acts on cut DNA molecules and joins their ends. This makes a new combination of circular autonomously replicating DNA created *in vitro* and is known as recombinant DNA. When this DNA is transferred into *Escherichia coli*, a bacterium closely related to *Salmonella*, it could replicate using the new host's DNA polymerase enzyme and make multiple copies. The ability to multiply copies of antibiotic resistance gene in *E. coli* was called **cloning** of antibiotic resistance gene in *E. coli*.

You can hence infer that there are three basic steps in genetically modifying an organism —

(i) identification of DNA with desirable genes;

(ii) introduction of the identified DNA into the host;

(iii) maintenance of introduced DNA in the host and transfer of the DNA to its progeny.

9.2 Tools of Recombinant DNA Technology

Now we know from the foregoing discussion that genetic engineering or recombinant DNA technology can be accomplished only if we have the key tools, i.e., restriction enzymes, polymerase enzymes, ligases, vectors and the host organism. Let us try to understand some of these in detail.

9.2.1 Restriction Enzymes

In the year 1963, the two enzymes responsible for restricting the growth of bacteriophage in *Escherichia coli* were isolated. One of these added methyl groups to DNA, while the other cut DNA. The later was called **restriction endonuclease**.

The first restriction endonuclease–*Hind II*, whose functioning depended on a specific DNA nucleotide sequence was isolated and characterised five years later. It was found that *Hind II* always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs. This specific base sequence is known as the **recognition sequence** for *Hind II*. Besides *Hind II*, today we know more than 900 restriction enzymes that have been isolated from over 230 strains of bacteria each of which recognise different recognition sequences.

The convention for naming these enzymes is the first letter of the name comes from the genus and the second two letters come from the species of the prokaryotic cell from which they were isolated, e.g., EcoRI comes from *Escherichia coli* RY 13. In EcoRI, the letter 'R' is derived from the name of

© NCERT not to be republished

strain. Roman numbers following the names indicate the order in which the enzymes were isolated from that strain of bacteria.

Restriction enzymes belong to a larger class of enzymes called **nucleases**. These are of two kinds; **exonucleases** and **endonucleases**. Exonucleases remove nucleotides from the ends of the DNA whereas, endonucleases make cuts at specific positions within the DNA.

Each restriction endonuclease functions by 'inspecting' the length of a DNA sequence. Once it finds its specific recognition sequence, it will bind to the DNA and cut each of the two strands of the double helix at specific points in their sugar-phosphate backbones (Figure 9.1). Each restriction endonuclease recognises a specific **palindromic nucleotide sequences** in the DNA.

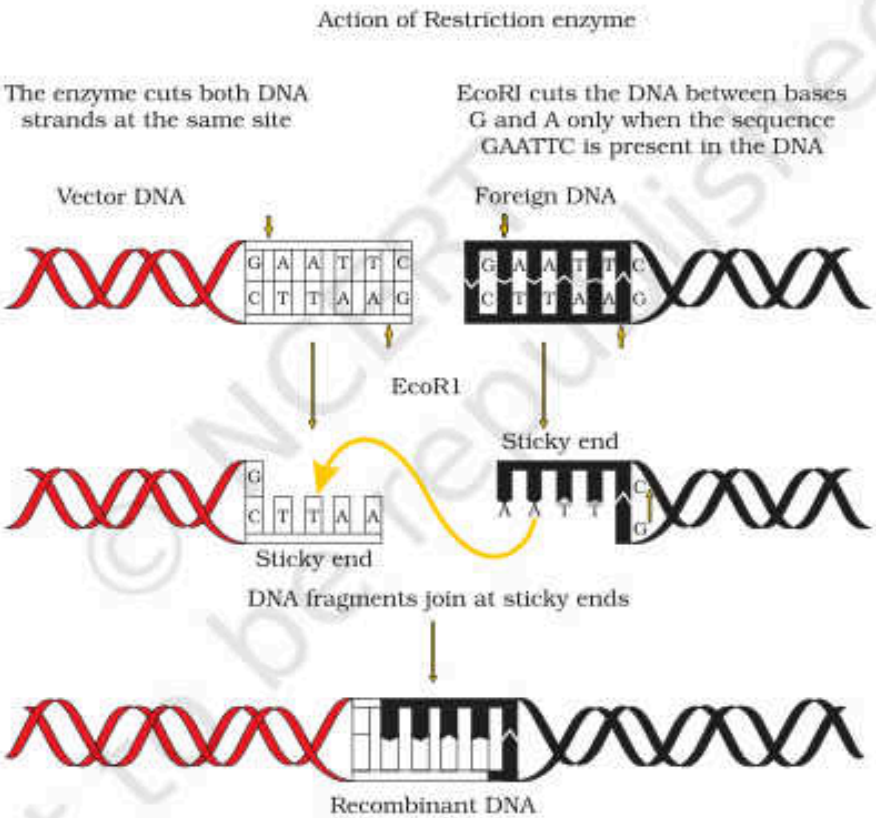


Figure 9.1 Steps in formation of recombinant DNA by action of restriction endonuclease enzyme - EcoRI

Do you know what palindromes are? These are groups of letters that form the same words when read both forward and backward, e.g., "MALAYALAM". As against a word-palindrome where the same word is read in both directions, the palindrome in DNA is a sequence of base pairs that reads same on the two strands when orientation of

© NCERT not to be republished